National Comprehensive Cancer Network®

2026

NCCN Guidelines for Patients®

Cancer care recommendations from leading experts at the National Comprehensive Cancer Network® (NCCN®)

Cervical Cancer



Presented with support from NCCN FOUNDATION® Guiding Treatment. Changing Lives.

Available online at NCCN.org/patientguidelines



NCCN Guidelines for Patients®

The essential guide for people facing cancer.

Based on care recommendations from leading cancer experts.

Explains high-quality cancer care provided at state-of-the-art cancer centers.

Reviewed and revised every year.

Did you know that top cancer centers across the United States work together to improve cancer care? This alliance of leading cancer centers is called the National Comprehensive Cancer Network® (NCCN®).

Because cancer care is always evolving, NCCN develops and frequently updates evidence-based cancer care recommendations used by health care providers worldwide. These recommendations are known as the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®).

The NCCN Guidelines for Patients plainly explain these expert recommendations, so you can talk with your care team about the best care for you.

These NCCN Guidelines for Patients are based on the NCCN Guidelines® for Cervical Cancer, Version 2.2026 — November 10, 2025.

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1
About cervical cancer

5 What is cervical cancer?

6 Is there more than one type?

7 Who is most at risk?

7 How is cervical cancer treated?

8 How can I get the best care?

Most cervical cancers are caused by human papillomavirus (HPV), a common sexually transmitted infection. The most common type of cervical cancer is squamous cell carcinoma (SCC).

The cervix is the narrow, lower end of the uterus. The uterus is where a baby grows and develops during pregnancy. During birth, the cervix opens and thins to allow the baby to move into the vagina.

What is cervical cancer?

Cervical cancer is a disease that starts when microscopic cells in the cervix undergo a change, usually for unknown reasons.

The (now abnormal) cells start constantly making copies of themselves. Over time, there are so many abnormal cells that a lump called a tumor develops.

Cancer cells can break off from the tumor and spread through your body in blood or in a bacteria-fighting fluid called lymph.

Your care team

Treatment for cervical cancer often involves a team of experts. Get to know your care team and don't be afraid to ask questions.



Is there more than one type?

Yes. The most common type of cervical cancer is **squamous cell carcinoma**. This type starts in the rounded part of the cervix that extends into the vagina, called the ectocervix.

Adenocarcinoma is the second most common type of cervical cancer. This type starts in the inner part of the cervix that forms a passage between the vagina and the uterus (the endocervix).

Adenocarcinomas are also often caused by human papillomavirus (HPV) and may be more difficult to detect than squamous cell carcinoma.

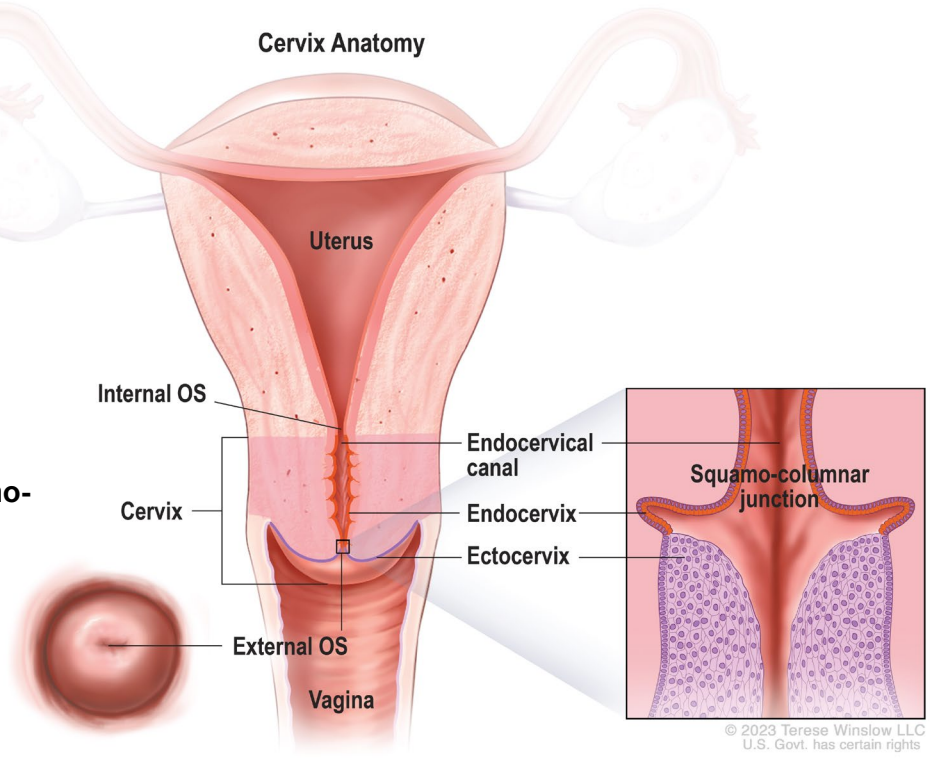
Less commonly, cervical cancers contain both squamous and adenocarcinoma cells. These are called **adenosquamous carcinomas** or mixed tumors.

The rarest and most aggressive type of cervical cancer is **neuroendocrine carcinoma of the cervix (NECC)**.

This guide doesn't discuss glassy cell carcinomas, sarcomas, or other uncommon types of cervical tumors.

The cervix

Most cervical cancers start in the area where the inner and outer cervix meet, called the transformation zone or squamo-columnar junction.



Who is most at risk?

Because almost all cervical cancers are caused by HPV, people who have had HPV are at highest risk. HPV is a common sexually transmitted infection.

In most people, the immune system gets rid of HPV from the body. In other people, the virus causes long-term cell changes that develop into cancer, often decades later. HPV also causes anal, head and neck, penile, vaginal, and vulvar cancers.

There are more than 100 strains (types) of HPV. Infection with certain strains is more likely to lead to cancer than others. Some strains can cause abnormal skin growths, called warts, to form on the anus, genitals, or other areas. A vaccine that protects against the highest-risk (and other) strains is available in the United States.

Other risk factors for cervical cancer fall into 2 general categories. They either increase the risk of being exposed to HPV, or they weaken the immune system, making it harder for the body to clear HPV infection.

How is cervical cancer treated?

The cancer stage is used to determine your options for treatment. The stage describes how far the cancer has (likely) spread, based on imaging and other testing.

When possible, treatment for early-stage tumors often involves surgery to remove the uterus and cervix (total hysterectomy). For some stage 1 cancers, fertility-sparing treatment may be an option, if desired.

For more advanced—but not metastatic—cervical cancers, treatment with platinum chemotherapy and radiation therapy (external and internal) is typically recommended. Information on these treatments is provided in Chapter 4.

For metastatic cancer, systemic therapy helps limit tumor growth and prevent further spread. Biomarker testing helps choose the best therapies for you. Local therapies can help relieve symptoms, such as pain and vaginal bleeding.

For a rare type of cervical cancer called small cell NECC, treatment often involves chemotherapy and radiation therapy (external and internal). Everyone with this uncommon cancer is encouraged to consider enrolling in a clinical trial, if available.

How can I get the best care?

Advocate for yourself. You have an important role to play in your care. Many people feel more satisfied when they actively take part in planning their cancer care.

The NCCN Guidelines for Patients will help you play a larger role in your care. Discuss the recommendations in this guide with your care team. Ask questions about your options and share your goals and concerns.

Don't know what to ask? You're not alone. That's why we include suggested questions to ask at the end of chapters.

Keep reading to find the best care for you.

How this guide can help you

Making decisions about cancer care is stressful. There's a lot to learn, and you don't know what the future holds.

Use this guide to get the information and support you need.

Patients, doctors, and other health care professionals trust the NCCN Guidelines for Patients. This guide uses clear, everyday language to explain current cancer care recommendations made by respected experts in the field. Their recommendations are based on the latest research and practices at leading cancer centers.

Your health is unique to you, so your cancer care should be, too. As you read this guide, you'll learn which treatments are likely to provide the best results for you. And you'll be better prepared to talk with your care team.

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Testing for cervical cancer

10 First steps

11 Cervical biopsy

11 Blood tests

12 Cone biopsy

13 Imaging

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15 Your care team

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16 Questions to ask

This chapter describes the tests used to diagnose cervical cancer and others needed to create your treatment plan.

First steps

If cervical cancer is suspected or confirmed, your doctor will do a complete physical exam. This often includes a pelvic exam to check the size and mobility of your uterus.

Your doctor will ask about your past and current health, and about symptoms that could be related to cervical cancer.

Some people with early cervical cancer don't have (or notice) any symptoms. Others may notice one or more of the following:

- Watery vaginal discharge
- Bleeding after sex
- Spotting (light bleeding between periods)

If you have symptoms of cervical cancer, the testing described in this chapter is recommended. Also see **Guide 1**.

Guide 1
Tests to diagnose cervical cancer and plan treatment

Cervical biopsy or cone biopsy
Blood tests: Complete blood count, blood chemistry profile, liver function tests, HIV test
Imaging: • Pelvic MRI to see the cervical tumor and check for cancer nearby • PET/CT (preferred) or CT to check for cancer in the liver, lungs, and abdomen
A check of your mental wellness and distress level
Tests to check for cancer in the bladder and bowel (if needed)
Help to quit smoking (if needed)
Family planning and options for preserving fertility (if needed)

Cervical biopsy

A cervical biopsy involves removing small samples of tissue from the cervix. It's the most commonly used procedure to diagnose cervical cancer. Your provider may remove tissue samples from the ectocervix, the endocervix, or both.

A pathologist examines the removed tissue. A pathologist is a doctor who specializes in evaluating cells and tissues under a microscope to diagnose disease. They look for areas of cancer or pre-cancer and determine the type of cervical cancer, when possible.

Blood tests

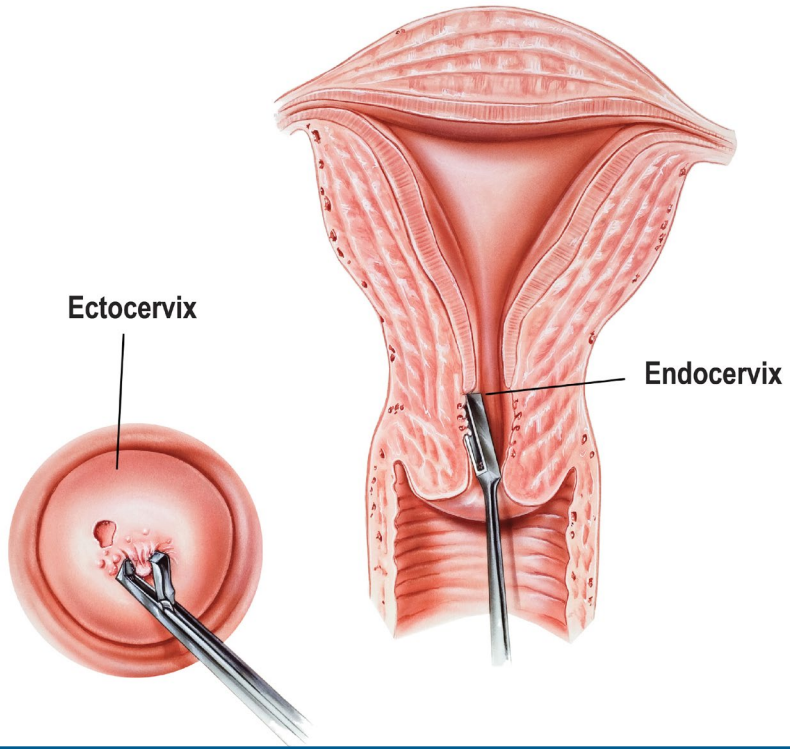
Testing for suspected cervical cancer typically includes the following blood tests. While they can't diagnose cancer, these tests provide a snapshot of your general health before treatment.

- Complete blood count (CBC)
- Blood chemistry profile
- Liver function tests

If you haven't been tested for human immunodeficiency virus (HIV) test, recently or ever, your doctor may recommend it. If you have HIV, talk to your provider about a referral to an HIV specialist. The treatment options described in this guide apply to people with or without HIV.

Cervical biopsy

A cervical biopsy is the most commonly used procedure to diagnose cervical cancer. A small sample of tissue is removed from the ectocervix, the endocervix, or both.



Cone biopsy

Cone biopsy involves removing a cone-shaped portion of the cervix. All of the transformation zone—where most cervical cancers start—is removed.

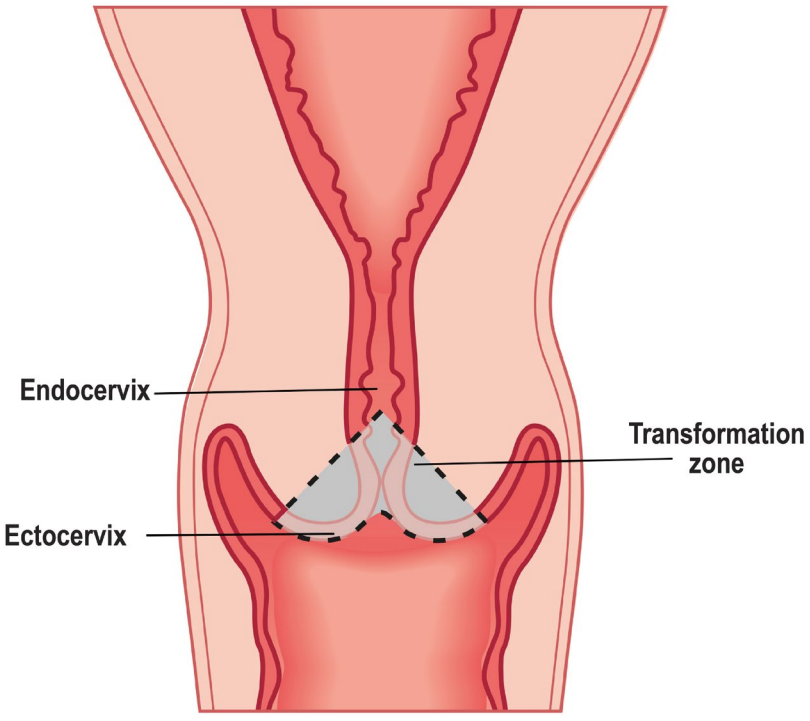
Your doctor may use a scalpel to remove the tissue. This method is called cold knife conization. Or they may use a thin loop of electrified (heated) wire. This is called loop electrosurgical excision procedure (LEEP).

After removing the cone-shaped sample of tissue, your doctor may use a tool called a curette to scrape a sample of tissue from deeper inside the cervical canal. This is called endocervical curettage.

The removed tissue samples are examined under a microscope by a pathologist.

Cone biopsy

A cone biopsy removes a cone-shaped section of the cervix that includes the transformation zone. Cone biopsy may be the only treatment needed for some early-stage cervical cancers.



Imaging

Imaging helps determine the exact location(s) and size of the cancer.

To make the images clearer, you will often receive a contrast agent. This dye-like substance may be put into your bloodstream through a vein, given to you as a drink, or both. Contrast causes some people to feel flushed or develop itchy, red spots on the skin.

Pelvis MRI

To see the cervical tumor and check for cancer nearby, MRI of the pelvis is preferred for most cervical cancers. MRI is good at showing if the cancer has spread to areas like the vagina, rectum, and bladder.

PET/CT

To see if cancer has spread to the liver, lungs, or other organs, PET, together with CT, is preferred for most cervical cancers. CT is a more detailed type of x-ray.

About an hour before the PET scan, you will be injected with a small amount of a radioactive substance, called a tracer. The tracer causes cancer cells to appear brighter than normal cells. The area from your neck to your groin will be scanned.

If you can't have PET, a CT of your chest, abdomen, and pelvis is recommended to look for cancer spread.

Pelvis MRI

An MRI of the pelvis is recommended as part of initial testing for cervical cancer. It can show the cervical tumor and nearby areas in detail. If you can't have MRI, transvaginal ultrasound is recommended.

MRI doesn't use radiation and is safe for most people. During the scan, you will lie face-up on a table that moves through a large tunnel-like machine.



Fertility and pregnancy

If you want the option of becoming pregnant in the future, talk to your care team about fertility-sparing treatment. It's typically only an option for small, early-stage cancers.

Also ask about seeing a reproductive endocrinologist. Reproductive endocrinologists are doctors that specialize in fertility. Although natural pregnancy won't be possible in most cases, some options to discuss are described below.

Ovarian transposition

Cervical cancer treatment may involve external beam radiation therapy (EBRT). Radiation damages the ovaries and causes them to stop producing the hormones needed for natural pregnancy. This can cause menopause to start early, called premature menopause.

Ovarian transposition is a surgery that moves one or both ovaries out of the range of the radiation beam. The medical name for this procedure is oophoropexy.

Ovarian transposition before starting EBRT may be an option if you are premenopausal and have the most common type of cervical cancer, squamous cell carcinoma.

Egg or embryo freezing

Unfertilized eggs can be removed, frozen, and stored for later use. Another name for this is oocyte preservation.

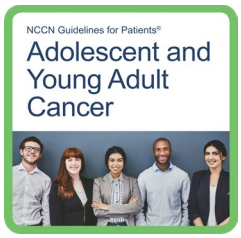
Surrogacy

If you have frozen eggs or frozen embryos, you may consider using a surrogate. A surrogate (often a relative or friend) volunteers to have the embryos inserted into their uterus. They carry the pregnancy and give birth.

More information

Recommendations for fertility-sparing treatment are provided in *Chapter 5: Treatment for common types*.

For more information on fertility preservation, see *NCCN Guidelines for Patients: Adolescent and Young Adult Cancer*, available at NCCN.org/patientguidelines and on the Patient Guides for Cancer app.



Your care team

Treatment for cervical cancer often involves a team of experts, including a gynecologic oncologist, a medical oncologist, and a radiation oncologist.

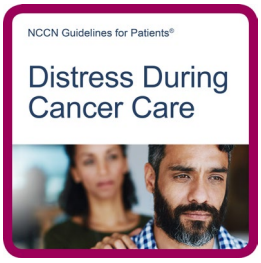
A gynecologic oncologist is an expert in surgery and chemotherapy for female reproductive cancers. A medical oncologist is an expert in treating cancer with systemic therapies (medicines), like chemotherapy. A radiation oncologist is an expert in the use of radiation therapy to treat cancer.

Other health care providers that you see for routine care can also be a part of your team. They can help you express your feelings about your care to the team. Treatment of other health problems may be improved if they are aware of and involved in your cancer care.

Distress

Depression, anxiety, fear, and distress are common after a cancer diagnosis. These feelings can make it harder to deal with cancer and its treatment. If you're worried, overwhelmed, or hopeless, ask your care team for help.

More information about cancer and distress is available at NCCN.org/patientguidelines and on the NCCN Patient Guides for Cancer app.



Key points

- Cervical cancer is most often diagnosed by cervical biopsy. Samples of cervical tissue are removed and tested for cancer and pre-cancer.
- A cone biopsy may be used to gather more information after a cervical biopsy, or to treat early cervical cancer. It involves removing a cone-shaped portion of the cervix.
- Blood tests provide helpful information about your general health and the health of your liver, kidneys, and other organs before treatment.
- To see the cervical tumor and check for cancer nearby, MRI of the pelvis is preferred for most cervical cancers.
- To see if cancer has spread to the liver, lungs, or other organs, PET/CT is preferred for most cervical cancers.
- A cystoscopy, proctoscopy, or both may be needed to look for cancer in the bladder and bowel.

Questions to ask

- Are tests for cervical cancer ever wrong?
- Should I get a second opinion about my diagnosis?
- What are the newest options to help me quit smoking?
- I want the option of becoming pregnant at some point. Will that be possible?



If you smoke, seek help to quit

Smoking increases the risk of lung problems during and after chemotherapy. It also increases your risk of developing other cancers. Learn about the strategies to help you quit in *NCCN Guidelines for Patients: Quitting Smoking,* available at NCCN.org/patientguidelines.

For online support, try these websites:

- SmokeFree.gov
- CDC.gov/tobacco

3
Staging

18 How is cervical cancer staged?

18 What are the stages?

28 Key points

28 Questions to ask

The results of imaging and other testing are used to determine the stage (extent) of the cancer. Your treatment options will depend on the cancer stage.

How is cervical cancer staged?

The following information is used to stage the cancer:

- The size or depth of the tumor
- Whether any lymph nodes have cancer
- Whether the cancer has spread to involve nearby organs, like the bladder or rectum
- Whether the cancer has spread to distant organs, such as the liver, lungs, or bone (metastasized)

What are the stages?

There are 4 main stages of cervical cancer: I (1), II (2), III (3), and IV (4). The stages are broken down into sub-stages that have letters and may also have a number.

The stages are described on the following pages. In general, people with earlier cancer stages have better outcomes, but not always.

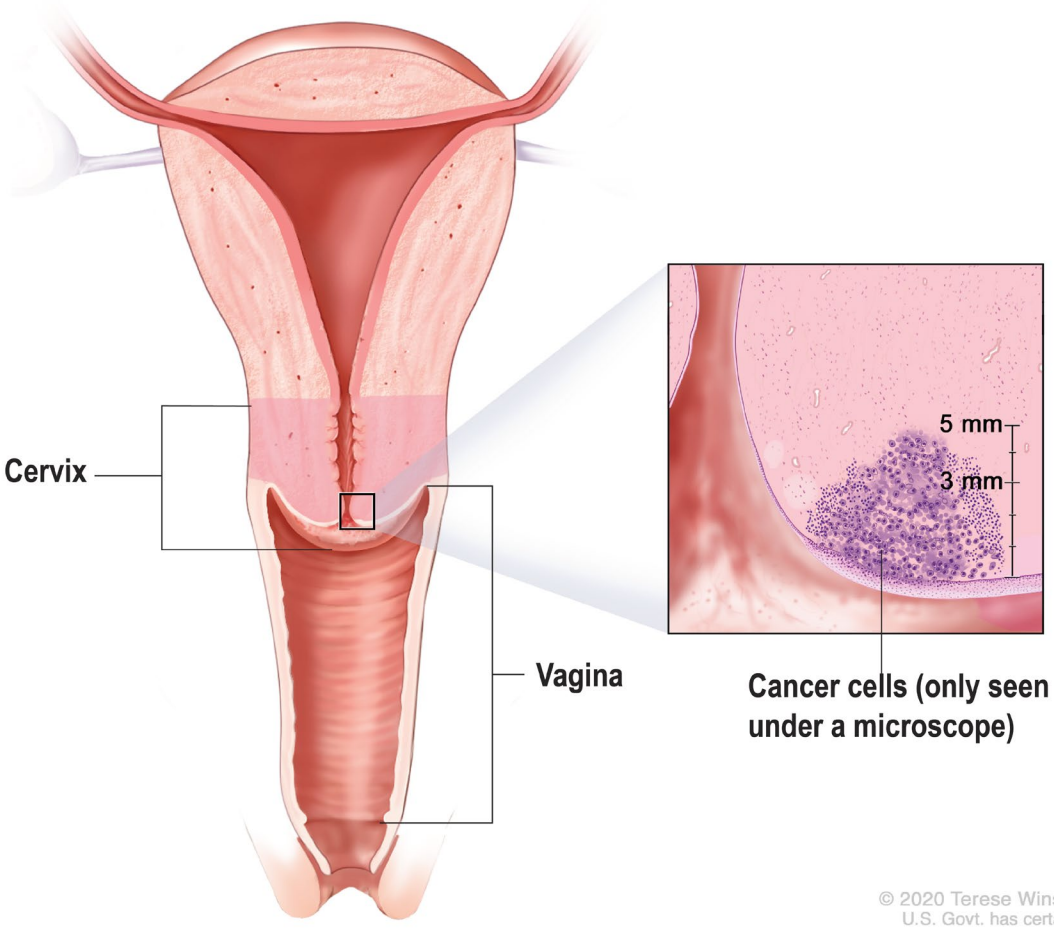
Treatment for cervical cancer takes a team of experts. The treatment that you and your care team agree on should be noted in the treatment plan, along with possible side effects.



Stage 1A cervical cancer

The cancer is 5 millimeters (mm) or smaller. 5 mm is about the size of a standard pencil eraser.

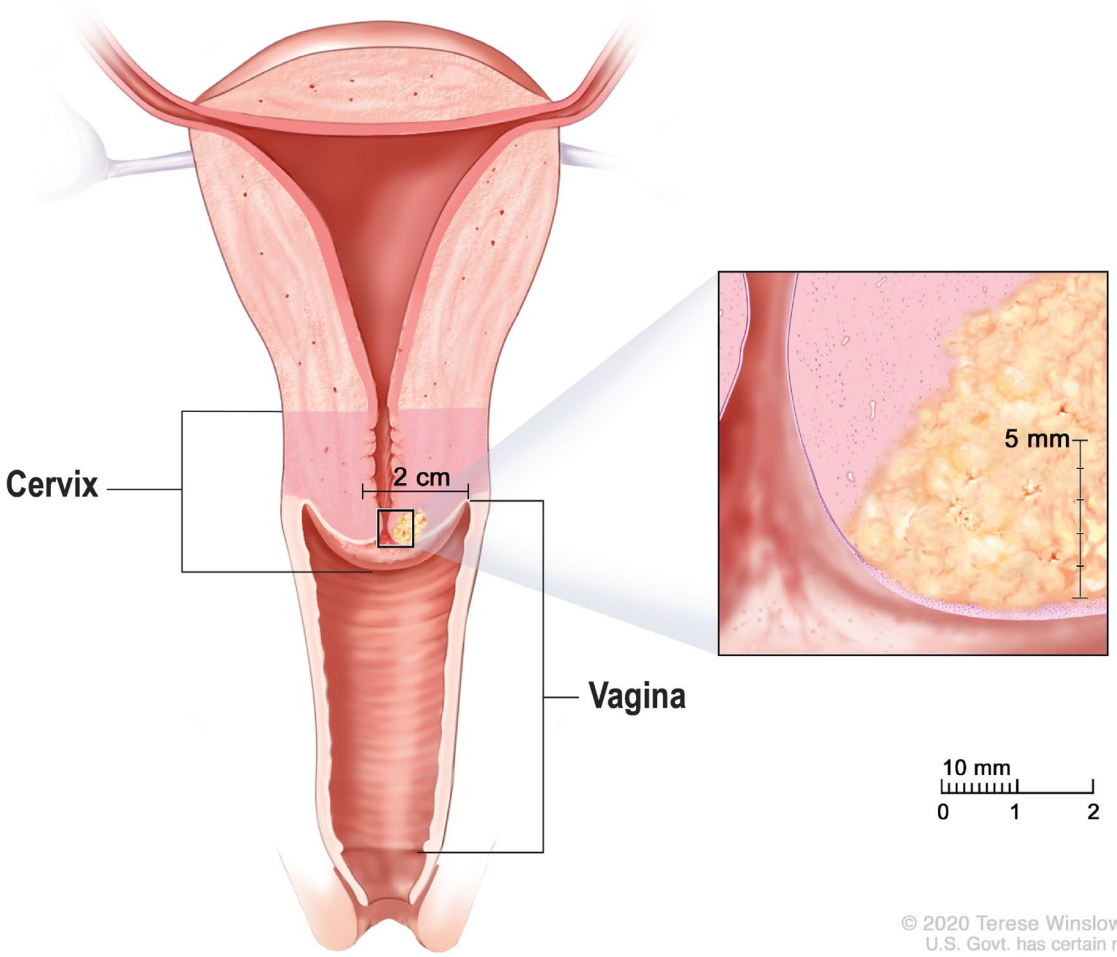
- **Cancers 3 mm or smaller are stage 1A1.**
- **Cancers between 3 and 5 mm are stage 1A2.**



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Stage 1B1 cervical cancer

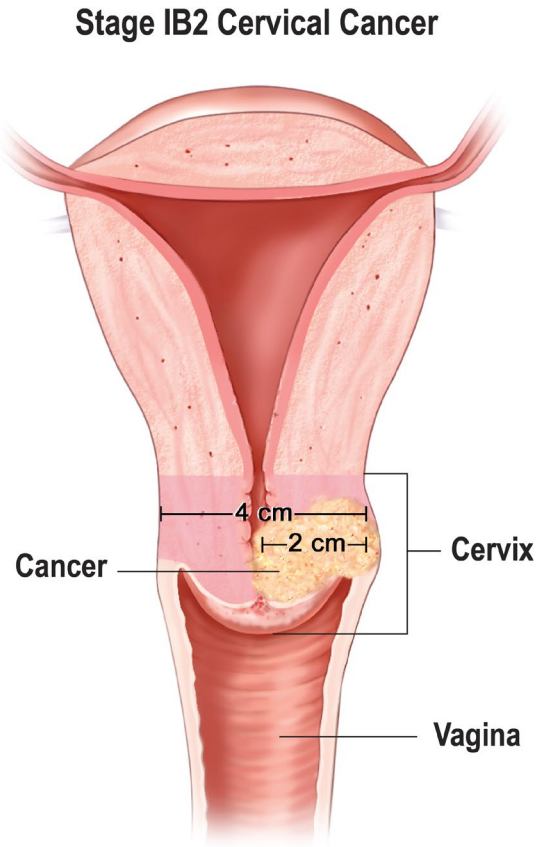
The cancer is larger than 5 mm but smaller than 2 cm.

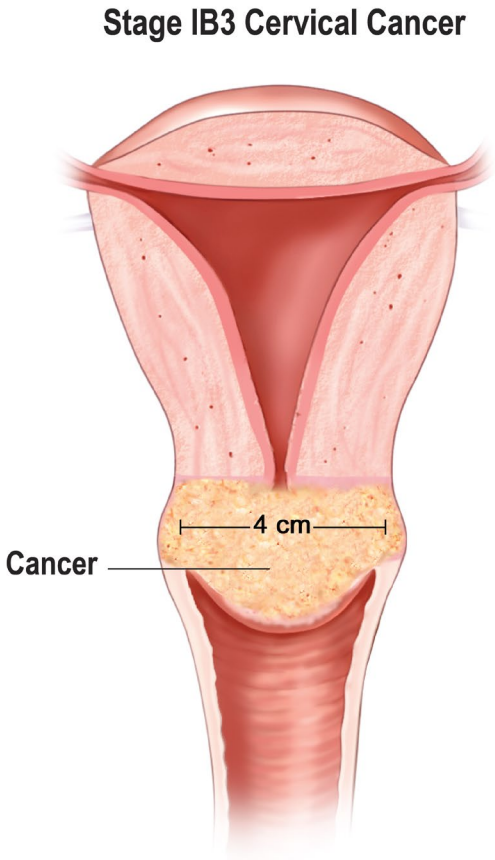


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Stage 1B2 and 1B3 cervical cancer

The cancer is only in the cervix. In stage 1B2, the cancer is between 2 and 4 cm. In stage 1B3, the cancer is larger than 4 cm.

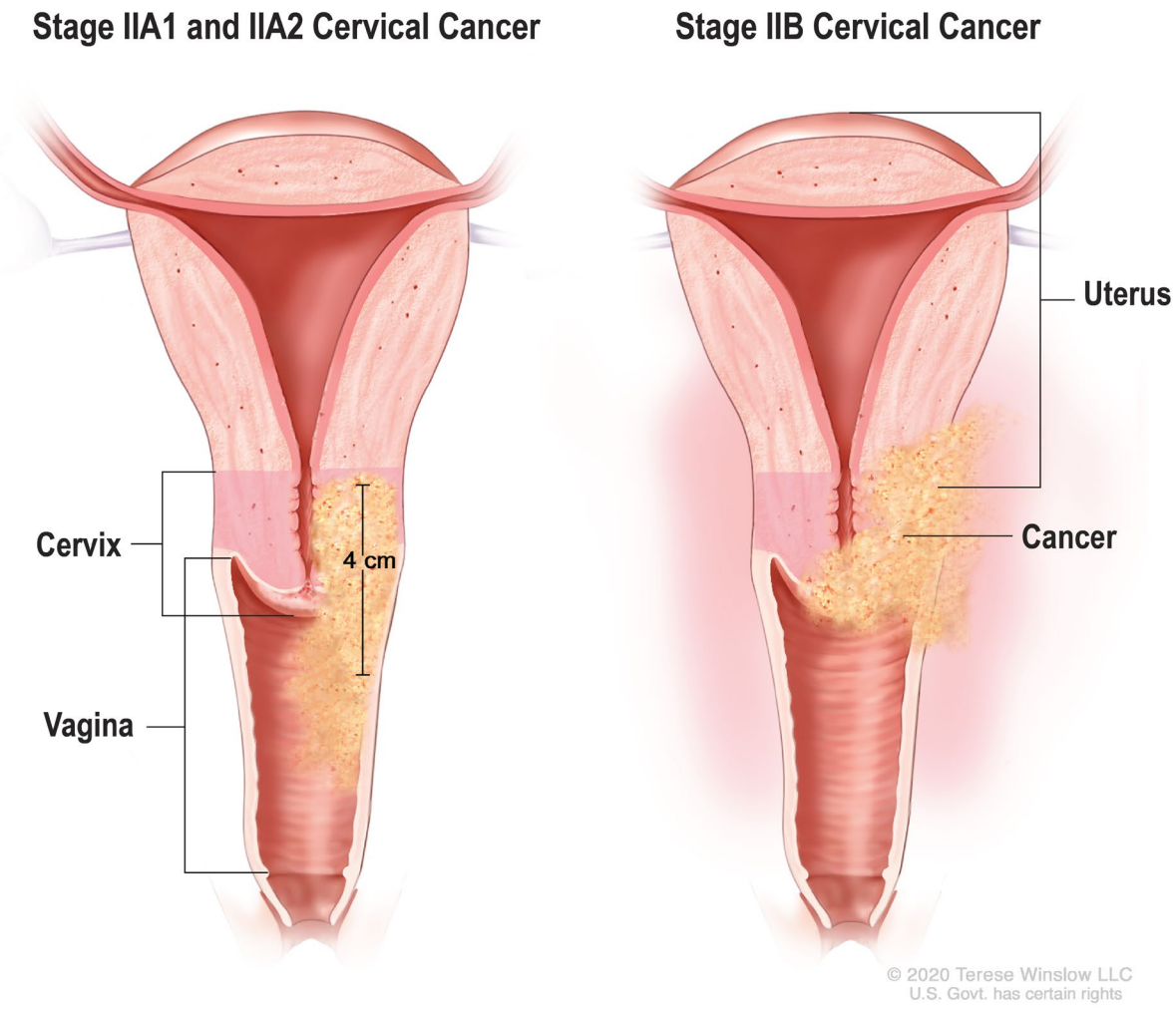




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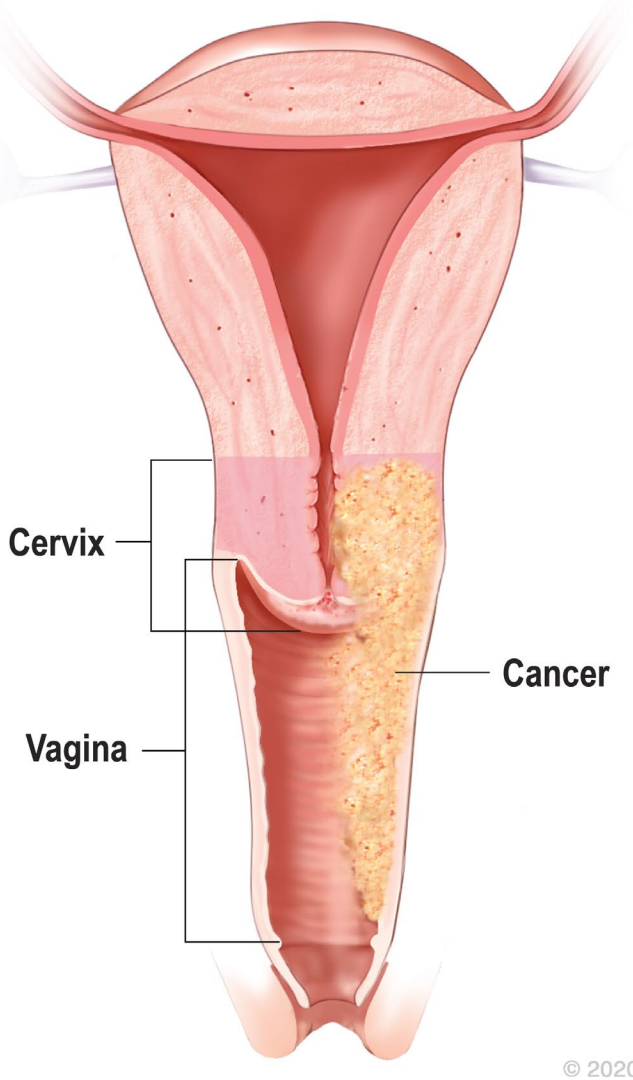
Stage 2 cervical cancer

The cancer has grown beyond the cervix. If the cancer has grown into the upper vagina, the stage is 2A. Stage 2A1 cancers are 4 cm or smaller. Stage 2A2 cancers are larger than 4 cm. Stage 2B cancer has grown into the parametrium. The parametrium is the fat and connective tissue that surrounds the cervix and uterus.



Stage 3A cervical cancer

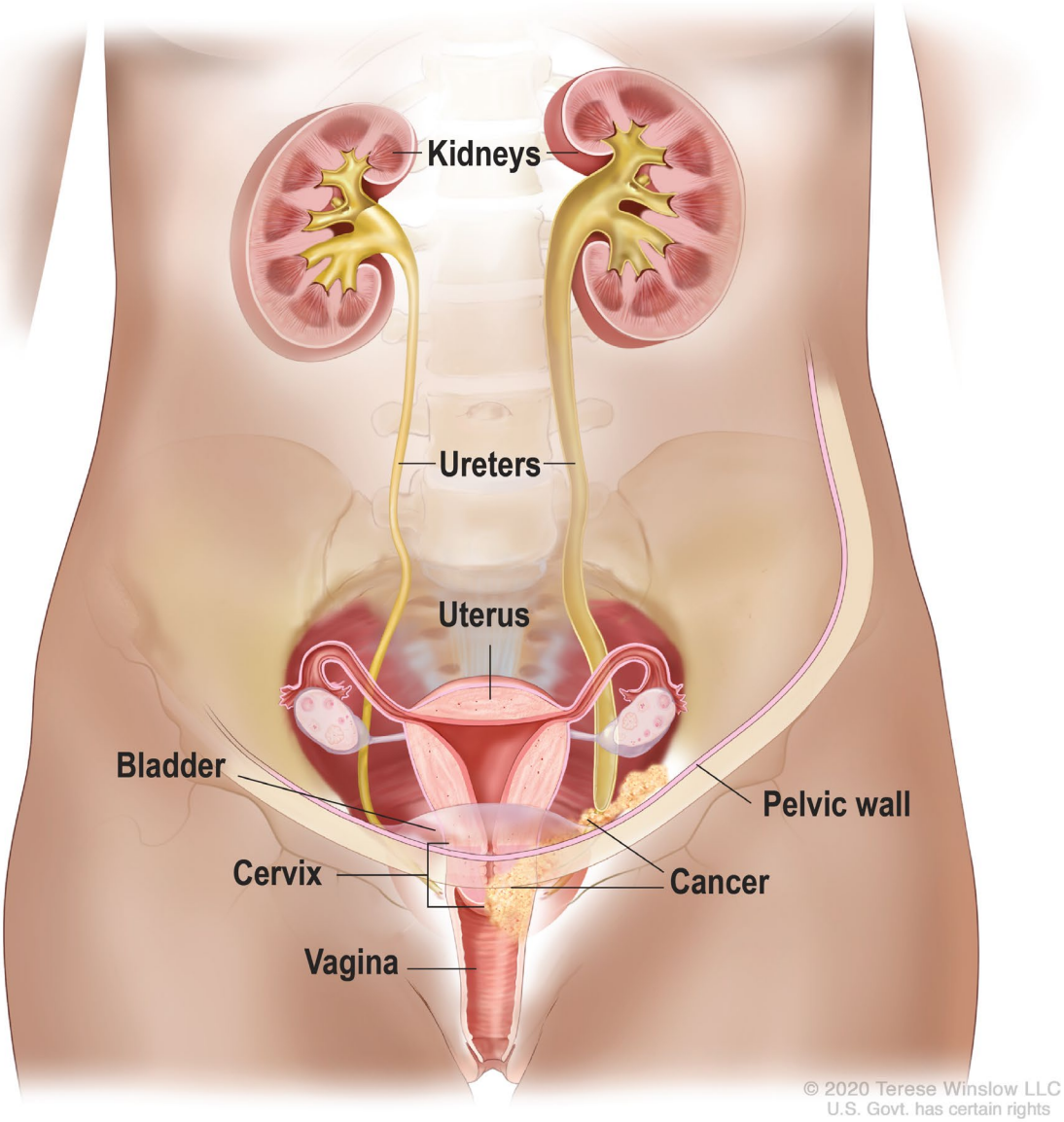
The cancer has grown into the lower third of the vagina.



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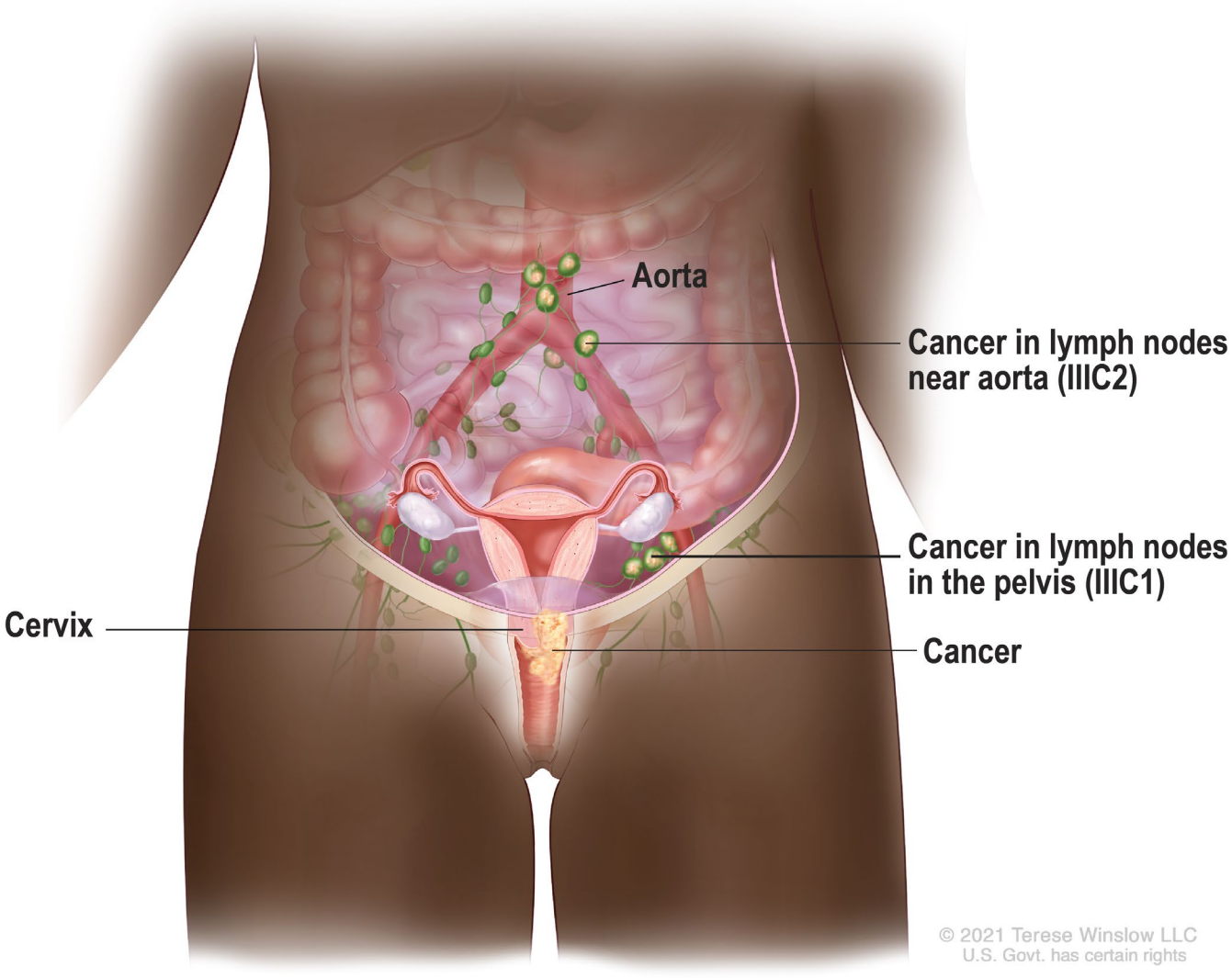
Stage 3B cervical cancer

The cancer has grown into the pelvic wall and/or has caused kidney swelling or dysfunction.



Stage 3C cervical cancer

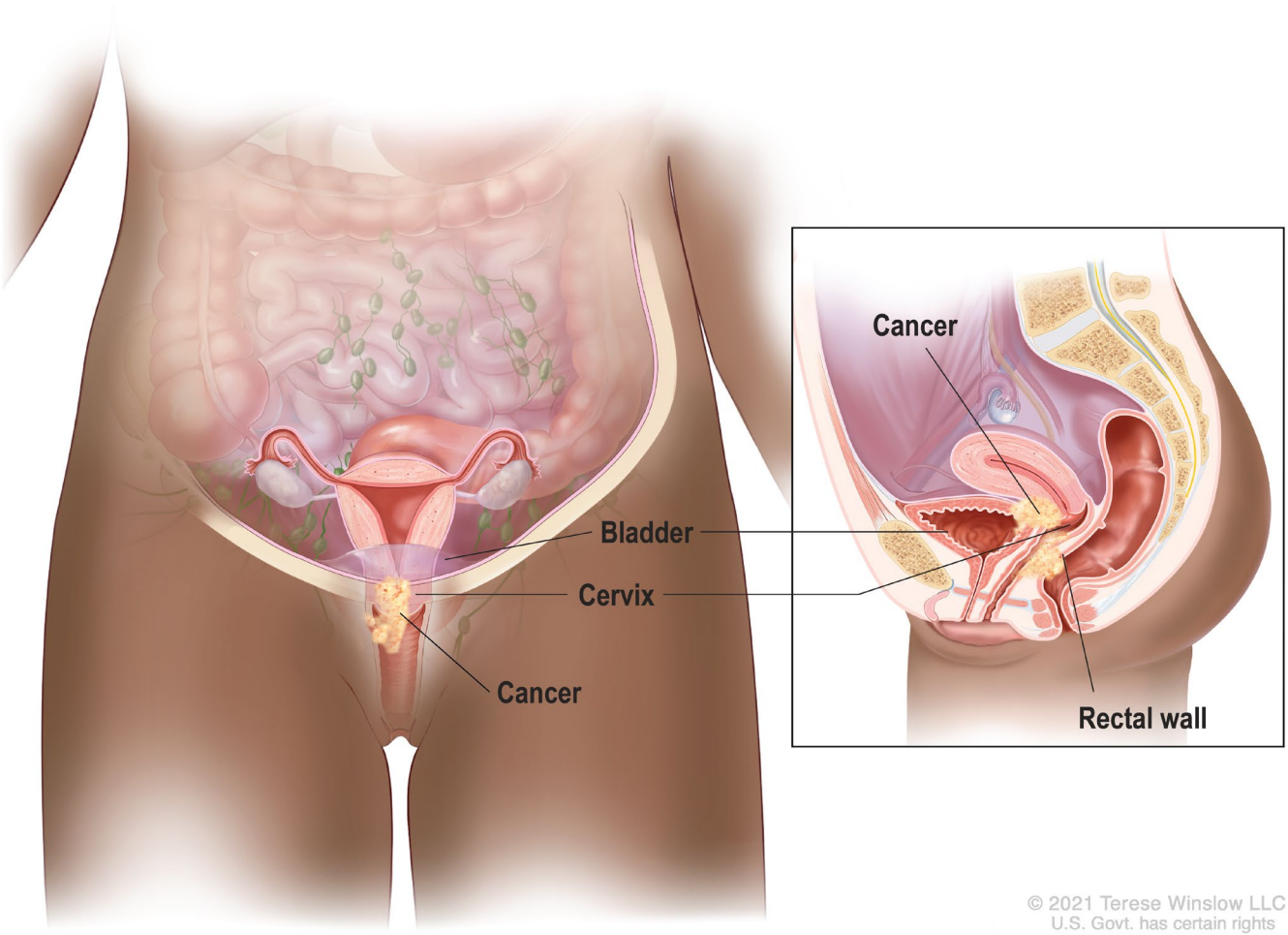
There is cancer in lymph nodes near the cervix (pelvic lymph nodes) and/or in lymph nodes in the abdomen, called the para-aortic lymph nodes.



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Stage 4A cervical cancer

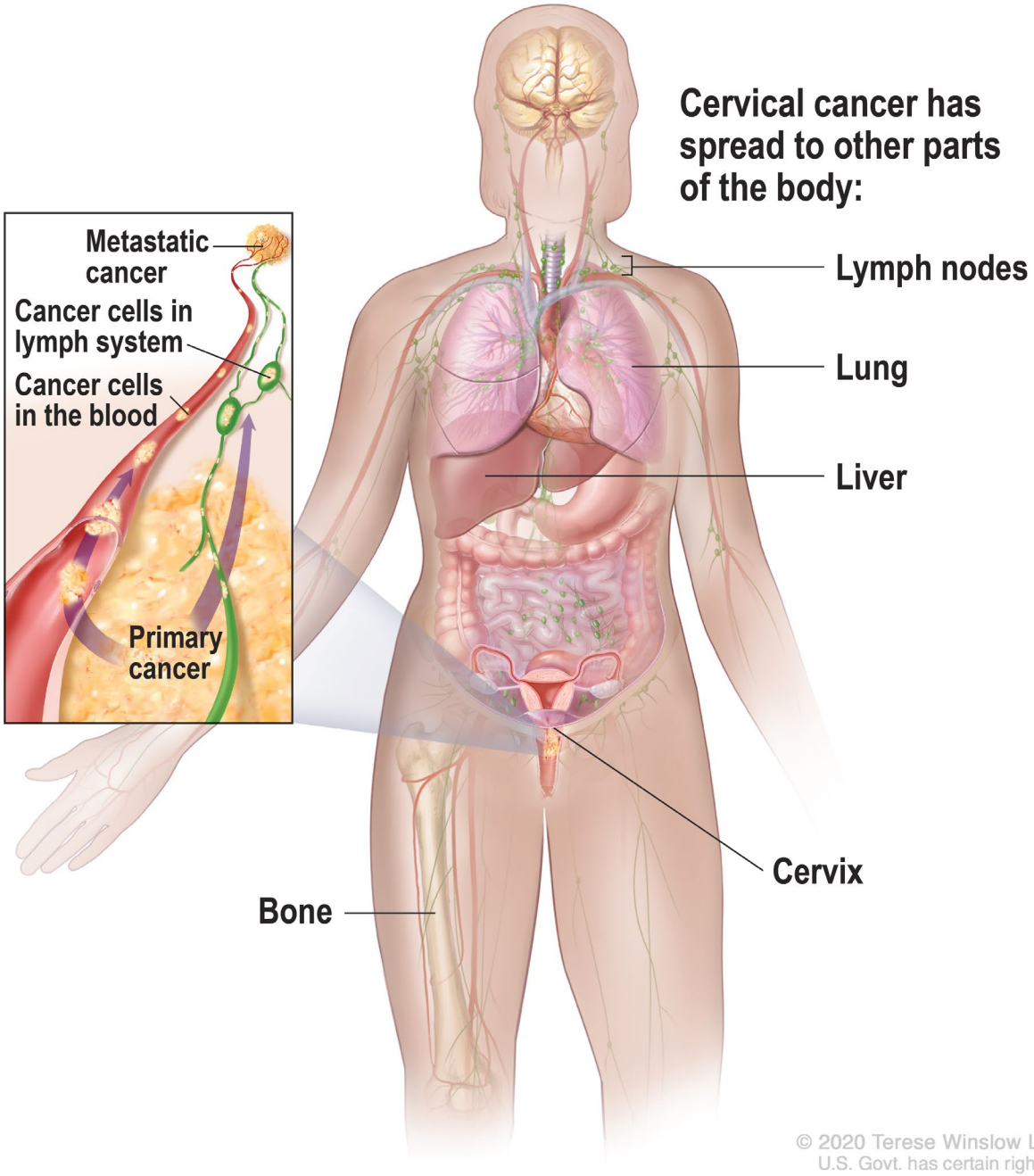
The cancer has spread to nearby organs, such as the bladder or rectum.



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Stage 4B cervical cancer

The cancer is metastatic. It has spread to the liver, lungs, abdomen, bone, or other distant sites or lymph nodes.



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Key points

- There are 4 main stages of cervical cancer: I (1), II (2), III (3), and IV (4).
- The stage describes how far the cancer has (likely) spread based on imaging and other testing.
- The cancer stage is used to determine your recommended options for treatment.

Questions to ask

- What stage is the cancer?
- Is it metastatic?
- If the cancer spreads, does the stage change?



Let us know what you think!

Please take a moment to complete an online survey about the NCCN Guidelines for Patients.
NCCN.org/patients/response

4
Types of treatment

30 Surgery

32 Radiation therapy

34 Chemoradiation

34 Systemic therapy

35 Clinical trials

36 Key points

36 Questions to ask

In this chapter, the main treatments for cervical cancer are described. Your treatment options will depend on the extent of the cancer and other factors.

Surgery

In most cases, surgery for cervical cancer involves removing the entire cervix. While the cervix can sometimes be removed through the vagina, a long incision (cut) through the abdomen is usually needed.

Minimally invasive (also called laparoscopic) surgery may be an option. This method involves making only a few small cuts into your body. Compared to surgery performed using a large cut through the abdomen, minimally invasive surgery usually causes less pain and scarring, and has a shorter recovery time.

Cone biopsy

Cone biopsy involves removing a cone-shaped section of the cervix that includes tissue from both the ectocervix and endocervix. It's a recommended treatment option for some small cervical cancers. See *Chapter 2: Testing for cervical cancer* for more information on cone biopsy.

Trachelectomy

Trachelectomy is surgery that removes the cervix, and sometimes part of the vagina. The uterus and ovaries are left in place, allowing for the possibility of natural pregnancy in the future.

In a simple trachelectomy, only the cervix is removed. In a radical trachelectomy, about 2 centimeters (a little less than an inch) of vaginal tissue is also removed. Both types can be performed through the vagina or through the abdomen.

Hysterectomy

Hysterectomy is surgery to remove the uterus (including the cervix). Depending on the type of hysterectomy performed, other nearby tissues may also be removed.

Type A (simple) hysterectomy removes only the uterus (including the cervix). It's most commonly used for very early cervical cancers. This type of hysterectomy can be performed through the vagina, through the abdomen, or using a minimally invasive approach.

Type B radical hysterectomy removes the uterus (including the cervix), some connective tissue that holds the cervix in place, and about a half inch or less of the vagina. This type of hysterectomy is performed through the abdomen.

Type C1 radical hysterectomy removes the uterus (including the cervix), much of the connective tissue that holds the cervix in place, and the top quarter or third of the vagina. This type of hysterectomy is performed through the abdomen.

There are other differences between these procedures. If surgery is planned, your surgeon will explain the procedure recommended for you in detail.

Will my ovaries be removed?

The ovaries may or may not be removed during hysterectomy. In premenopausal people, surgically removing the ovaries causes a sudden drop in estrogen in the body. This is called surgical menopause.

Symptoms include hot flashes, sleeping problems, night sweats, weight gain, and changes in mood. It's also common for the lining of the vagina to become thin, dry, and inflamed. This is called vaginal atrophy.

Long-term risks of not having enough estrogen include heart disease and weakened bones (osteoporosis). If hysterectomy is being considered, ask your provider about the benefits and risks of keeping your ovaries.

Checking lymph nodes for cancer

Lymph nodes are bean-shaped glands that help the body fight infection and disease. During surgery for cervical cancer, lymph nodes may be removed in order to be tested for cancer. This is called lymph node dissection or lymphadenectomy.

Testing the removed lymph nodes helps determine the extent (spread) of the cancer. Removing the nodes can also help limit the spread of cancer cells through lymph.

The lymph nodes closest to the cervix are the pelvic nodes. Cervical cancer generally spreads to these nodes first. Lymph nodes in the abdomen, called the para-aortic nodes, are also sometimes removed.

To identify and remove the lymph node(s) most likely to contain cancer, a sentinel lymph node biopsy may be performed. This involves injecting a special dye or a radioactive substance into the cervix near the cancer. Lymph nodes containing the dye or substance can be seen using a special camera. These nodes are at the highest risk of cancer spread and are called the sentinel nodes.

Staging before surgery

For cancers too advanced to be removed by hysterectomy, lymph nodes may be removed from your pelvis or abdomen and tested before any treatment is given. Knowing whether the nodes contain cancer helps your provider make the most informed recommendations for your care.

A minimally invasive approach is typically used to access and remove the lymph nodes. This is called laparoscopic surgical staging.

Radiation therapy

The 2 most common types of radiation therapy used to treat cervical cancer are external beam radiation therapy (EBRT) and brachytherapy.

External beam radiation therapy

In EBRT, a large machine aims radiation at the cancer site. The radiation passes through skin and other tissue to reach the tumor and nearby lymph nodes. A planning session, called simulation, is needed before treatment begins. Ask your care team what to expect.

The type(s) of EBRT you may receive depends on the area of your body being treated. Intensity-modulated radiation therapy (IMRT) delivers a high dose of radiation to the cervical tumor while limiting the amount that reaches surrounding normal tissue and nearby organs. This helps reduce treatment-related side effects.

Stereotactic body radiation therapy (SBRT) may be used to treat metastatic cervical cancer that has spread to the liver or lungs. This method usually only requires 5 treatment sessions.

Early menopause

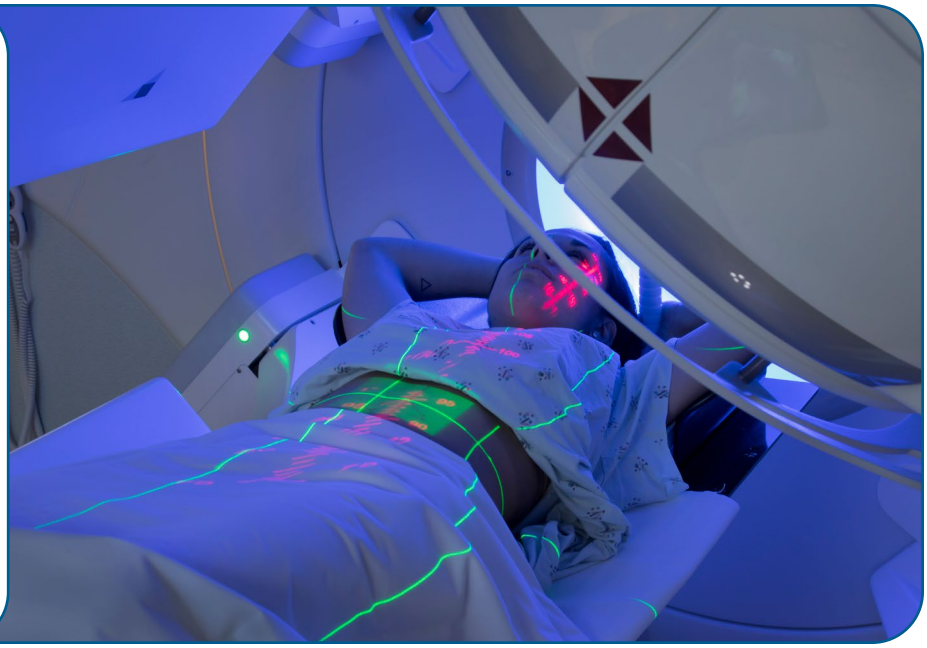
When the ovaries are included in the radiation field, it causes premature (early) menopause. The symptoms are similar to those caused by surgical menopause.

Ovarian transposition is a surgery that moves one or both ovaries out of the range of the radiation beam. This procedure may be an option if you are premenopausal and the cancer is squamous cell carcinoma.

Otherwise, your provider may suggest the use of menopausal hormone therapy (MHT) after radiation therapy has completed. Hormone therapy for menopause helps lessen some of the side effects of radiation-induced menopause. This approach used to be called hormone replacement therapy, or HRT.

External beam radiation therapy

Common side effects during the 5 to 6 weeks of EBRT include fatigue, skin redness and irritation, diarrhea, nausea, and frequent or painful urination.



Radiation therapy for cervical cancer can also have long-term and serious side effects on fertility, sexual health, and bowel and bladder function.

See *Chapter 7: Survivorship* for more information on the possible long-term risks.

Brachytherapy

Also known as internal radiation therapy, brachytherapy involves treatment with radioactive material placed inside the body. Brachytherapy allows a high dose of radiation to be targeted at the tumor while limiting the amount delivered to surrounding normal tissue. Iti's critical for the treatment of some cervical cancers, because there is a limit to how much external radiation can be used.

During brachytherapy, instruments are placed into the cervix, uterus, and vagina. A radioactive material then travels into the these instruments. This method is known as intracavitary brachytherapy. Sometimes, additional instruments (catheters, or thin hollow tubes) are placed in the tumor itself or in tissues next to the tumor (interstitial brachytherapy).

For the treatment of cervical cancer, intracavitary brachytherapy is most often used. Interstitial brachytherapy is a specialized technique that is used for more advanced cancers that involve the parametrium. It is best performed at cancer centers with experience in this method.

Brachytherapy for cervical cancer may be given in short bursts, called high dose-rate (HDR) brachytherapy, or in long bursts, called low dose-rate (LDR) brachytherapy. Most centers currently use HDR brachytherapy.

Treatment is typically given in 3 to 5 sessions over 2 weeks.

For those receiving treatment primarily with radiation and systemic therapy, a device called an applicator is inserted into the uterus through the vagina. A tandem and ring applicator is commonly used for HDR brachytherapy. The tandem is a long, thin tube that extends into the uterus. The ring is a hollow circle that stays in the vagina, pressed against the cervix. A tandem and ovoid applicator uses hollow, rounded capsules instead of a ring. General anesthesia or deep sedation is often needed, especially for the first applicator placement.

The applicator is placed to align with the tissue targeted for treatment, often with the help of an imaging technique (MRI or CT). These images are used to design the brachytherapy treatment plan. The applicator is connected to a brachytherapy machine. A radiation source travels from the machine through the hollow tubes and into the applicator. Treatment takes about 10 minutes.

Sometimes brachytherapy is given as treatments over a couple of days. In this case, the instruments are placed and left in the body until all the treatments have been given (1 to 2 days). You will need to remain in bed until treatment is complete so that the instruments don’t move.

Brachytherapy is typically started in the final week or right after completing EBRT. Shrinking the tumor with EBRT first allows for better placement of the brachytherapy applicators. Treatment with both types of radiation therapy should ideally be completed within 7 to 8 weeks.

Chemoradiation

EBRT and chemotherapy are often used together during the same time period to treat cervical cancer. This treatment strategy is called chemoradiation. Chemoradiation is recommended for most locally advanced cervical cancers. These cancers have grown beyond the cervix but haven't spread to the liver, lungs, or bones.

As part of chemoradiation, there are typically 5 EBRT treatment sessions per week for 5 to 6 weeks. Chemotherapy with cisplatin is typically given on a 7-day cycle (once weekly) during this time. An immunotherapy treatment called pembrolizumab (Keytruda) may also be given with chemotherapy. Immunotherapy boosts your immune system to better fight the cancer.

Systemic therapy

Systemic therapy is treatment with substances that travel in the bloodstream, reaching cells throughout the body. Chemotherapy, targeted therapy, and immunotherapy are types of systemic therapy.

Platinum chemotherapy is the most commonly used systemic therapy for cervical cancer. Cisplatin is generally preferred, but carboplatin may be used instead if needed. Targeted therapy and immunotherapy are newer types of systemic therapy.

Most systemic therapies are slowly infused into your bloodstream through a vein. Infusions are often given in cycles of treatment days followed by days of rest. This allows your body to recover between cycles.

Common side effects of systemic therapy include nausea, vomiting, and mouth sores. Ask your care team for a list of all possible side effects. Cisplatin can damage the kidneys and can also cause ringing in the ears and hearing problems.

Levels of preference

All treatments listed in this guide are recommended and appropriate. When helpful, NCCN experts also assign a level of preference to their recommendations for systemic therapies:

- **Preferred therapies** have the most evidence they may work better and be safer than other therapies.
- **Other recommended therapies** can provide effective results but may have less evidence, more side effects, or may not work quite as well as preferred therapies.
- **Therapies used in certain cases** work best for individuals with specific cancer features or health circumstances.

Clinical trials

A clinical trial is a type of medical research study. After being developed and tested in a lab, potential new ways of fighting cancer need to be studied in people.

If found to be safe and effective in a clinical trial, a drug, device, or treatment approach may be approved by the U.S. FDA.

Everyone with cancer should carefully consider all of the treatment options available for their cancer type, including standard treatments and clinical trials. Talk to your doctor about whether a clinical trial may make sense for you.

Phases

Most cancer clinical trials focus on treatment and are done in phases.

- **Phase 1** trials study the safety and side effects of an investigational drug or treatment approach.
- **Phase 2** trials study how well the drug or approach works against a specific type of cancer.
- **Phase 3** trials test the drug or approach against a standard treatment. If the results are good, it may be approved by the FDA.
- **Phase 4** trials study the safety and benefit of an FDA-approved treatment.

Who can enroll?

It depends on the clinical trial's rules, called eligibility criteria. The rules may be about age, cancer type and stage, treatment history, or general health. They ensure that participants are alike in specific ways and that the trial is as safe as possible for the participants.

Informed consent

Clinical trials are managed by a research team. This group of experts will review the study with you in detail, including its purpose and the risks and benefits of joining. All of this information is also provided in an informed consent form. Read the form carefully and



Finding a clinical trial

In the United States

NCCN Cancer Centers
NCCN.org/cancercenters

The National Cancer Institute (NCI)
cancer.gov/about-cancer/treatment/clinical-trials/search

Worldwide

The U.S. National Library of Medicine (NLM)
clinicaltrials.gov

Need help finding a clinical trial?

NCI's Cancer Information Service (CIS)
1.800.4.CANCER (1.800.422.6237)
cancer.gov/contact

ask questions before signing it. Take time to discuss it with people you trust. Keep in mind that you can leave and seek treatment outside of the clinical trial at any time.

Will I get a placebo?

Placebos (inactive versions of real medicines) are almost never used alone in cancer clinical trials. It's common to receive either a placebo with a standard treatment, or a new drug with a standard treatment. You will be informed, verbally and in writing, if a placebo is part of a clinical trial before you enroll.

Are clinical trials free?

There is no fee to enroll in a clinical trial. The study sponsor pays for research-related costs, including the study drug. But you may need to pay for other services, like transportation or childcare, due to extra appointments. During the trial, you will continue to receive standard cancer care. This care is often covered by insurance.

Key points

- Hysterectomy is surgery that removes the uterus, including the cervix. Some of the vagina and cervical connective tissue may also be removed. The ovaries may be removed or left in place. Removing them causes early (also called surgical) menopause.
- Trachelectomy is surgery that removes the cervix. It's a type of fertility-sparing surgery. The upper part of the vagina may also be removed.
- Radiation therapy uses high-energy x-rays to kill cancer cells. In external beam radiation therapy (EBRT), a large machine aims radiation at the cancer site.
- Brachytherapy involves treatment with radioactive material placed inside the body. It delivers a high dose of radiation directly to the tumor.
- Platinum chemotherapy is the most commonly used systemic therapy for cervical cancer.
- EBRT and platinum chemotherapy given during the same time period is a treatment strategy called chemoradiation.
- Clinical trials provide access to investigational treatments that may, in time, be approved by the FDA.

Questions to ask

- What are the side effects of systemic therapy?
- What are the side effects of EBRT? Of brachytherapy?
- What happens during the simulation session for EBRT?
- Should I consider a clinical trial?

5
Treatment for common types

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This chapter explains treatment for common types of cervical cancer. Surgery is usually recommended for early-stage cancers, while locally advanced cancers are often treated with chemoradiation.

This section applies to the following types of cervical cancer:

- Squamous cell carcinoma
- Adenocarcinoma
- Adenosquamous carcinoma

Treatment is based on the cancer stage. The stage describes how far the cancer has probably spread based on imaging and other testing.

A risk factor that guides treatment for cervical cancer is whether there are cancer cells in blood vessels or lymph vessels near the tumor. This is called **lymphovascular space invasion (LVSI)**.

Cancers with LVSI are more likely to spread to nearby lymph nodes. Removing and testing lymph nodes near the cancer will be part of your care.

Fertility-sparing treatment for early-stage cancer

Surgical treatment for early-stage cervical cancer often involves surgery to remove the uterus (hysterectomy). Carrying a pregnancy isn't possible afterward. If the cancer hasn't spread beyond the cervix, fertility-sparing treatment may be possible.

This approach involves surgery that doesn't remove the uterus or ovaries, allowing you the option of natural pregnancy in the future. On the next page, fertility-sparing treatment for stage 1A1, 1A2, and 1B1 cancers is explained.

If you are considering fertility-sparing treatment, ask your provider about getting the opinion of a reproductive endocrinologist. These are doctors that specialize in fertility.



Stage 1A1

Cone biopsy is recommended for all stage 1A1 cancers. If the cancer is known to have LVSI, your surgeon will also remove nearby lymph nodes for testing.

If the results of cone biopsy are good, you don't need more treatment. If the results suggest that the cancer wasn't fully removed, there are 2 possibilities.: Your doctor may recommend another cone biopsy, or a trachelectomy.

Stage 1A2

Cone biopsy with lymph node removal is recommended for all stage 1A2 cancers. Your surgeon will remove nearby lymph nodes in the pelvis for testing.

If cancer is found in any of the lymph nodes removed during surgery, chemoradiation is recommended. Vaginal brachytherapy may be paired with chemoradiation. See *Chapter 4: Types of treatment* for information on these treatments.

Stage 1B1

Cone biopsy is recommended for low-risk stage 1B1 cancers. Low-risk generally means that there is no LVSI and the cancer meets other low-risk criteria. Your surgeon will also remove nearby lymph nodes in the pelvis for testing.

For stage 1B1 cancers that don't meet the criteria for cone biopsy, radical trachelectomy is recommended. Your surgeon will also remove lymph nodes in the pelvis, and possibly in the abdomen, for testing.

If no cancer is found in the removed lymph nodes or in tissues outside the cervix, that is good news. But you may still need external beam radiation therapy (EBRT) next. Your doctor will consider the tumor size and other features to determine if EBRT would be helpful. If you don't need EBRT, observation will begin.

If cancer is found in any of the lymph nodes removed during surgery, or in any tissues outside the cervix, chemoradiation is recommended. Vaginal brachytherapy may be paired with chemoradiation.

Stage 1B2

Fertility-sparing surgery isn't performed often for stage 1B2 cancers. Abdominal surgery is generally preferred.

Treatment for early-stage cancer (non–fertility-sparing)

This section describes non–fertility-sparing treatment for stage 1 and stage 2A cancers. Stage 1 cancer is only in the cervix. Stage 2A cancers have spread to the upper vagina.

Stage 1A1

Stage 1A1 cancers are typically diagnosed by cone biopsy. Your next treatment depends on the cone biopsy results, and whether hysterectomy can be safely performed. **See Guide 2**.

Guide 2
Stage 1A1: Non–fertility-sparing treatment after cone biopsy

Cone biopsy results	Type of hysterectomy recommended	Non-surgical option(s) if needed or desired
No cancer or pre-cancer in margins	Type A hysterectomy (and lymph node removal, for cancers with LVSI)	For cancers without LVSI: • Observation For cancers with LVSI: • EBRT plus brachytherapy
Dysplasia (pre-cancer) in margins	Type A hysterectomy (and lymph node removal, for cancers with LVSI)	EBRT plus brachytherapy
Carcinoma (cancer) in margins	Type B radical hysterectomy and lymph node removal	For cancers without LVSI: • Another cone biopsy **(preferred)** • Brachytherapy, with or without EBRT For cancers with LVSI: • EBRT plus brachytherapy

Stages 1A2 and 1B1

Type A hysterectomy is recommended for low-risk stage 1A2 and 1B1 cancers. Low-risk generally means that there is no LVSI and the cancer meets other low-risk criteria. Your surgeon will also remove nearby lymph nodes for testing.

For all other stage 1A2 and 1B1 cancers, a type C1 radical hysterectomy is recommended. The surgeon will remove lymph nodes from the pelvis, and maybe from the abdomen, during surgery.

If surgery isn't an option, treatment with EBRT and brachytherapy is recommended. Platinum chemotherapy might also be given during the same time period.

Stage 1B2 and 2A1

A type C1 radical hysterectomy is recommended for stage 1B2 and 2A1 cancers. The surgeon will remove lymph nodes from the pelvis, and maybe from the abdomen, during surgery.

If surgery isn't an option, treatment with EBRT and brachytherapy is recommended. Platinum chemotherapy might also be given during the same time period.

Stage 1B3 and 2A2

These cancers may be treated as early-stage (with surgery) or as locally advanced (with chemoradiation and brachytherapy). If treatment with chemoradiation and brachytherapy is planned, you may have chemotherapy first. This is called induction chemotherapy.

If surgery is planned, a type C1 radical hysterectomy is recommended. The surgeon will remove lymph nodes from your pelvis, and maybe from your abdomen.

Do I need more treatment?

After surgery, the pathologic (surgical) stage of the cancer is determined by examining the removed tissue. For stage 1 and stage 2A cancers, possible next steps are described below.

If testing finds no cancer in tissue beyond the cervix or in lymph nodes removed during surgery, you may still need more treatment. EBRT is recommended for cancers with risk factors. Your provider will consider the tumor size, whether there is LVSI, and other factors to determine if EBRT is a good choice for you.

If EBRT is planned, platinum chemotherapy may be given during the same time period. This is called chemoradiation.

If cancer is found in any of the removed lymph nodes, or in any tissues outside the cervix, treatment with chemoradiation is recommended. Vaginal brachytherapy may be given in addition to EBRT and platinum chemotherapy.

It's important to take care of yourself during cancer care. Try to be active, eat nutritious foods, drink plenty of fluids, and do things that make you feel energized.



Treatment for locally advanced cancer

Cervical cancer that has grown beyond the cervix but hasn't spread to the liver, lungs, or bones is described as locally advanced. Locally advanced cancers are not metastatic.

The information in this section applies to the following stages:

- Stage 2B
- Stage 3
- Stage 4A

Checking lymph nodes

Locally advanced cancers are more likely to have spread to lymph nodes than early-stage cancers. You may have extra imaging to look for lymph nodes with cancer.

Another approach is to surgically remove and test lymph nodes before any treatment is given. A minimally invasive method is typically used to access and remove the lymph nodes. This is called surgical staging.

If surgical staging is performed, lymph nodes in the abdomen (para-aortic nodes) are typically removed. Pelvic lymph nodes may or may not be removed.

Treatment

Treatment with chemoradiation and brachytherapy is recommended for most locally advanced cervical cancers. Depending on the cancer stage, pembrolizumab (Keytruda) may also be given with the platinum chemotherapy you receive for chemoradiation.

Before chemoradiation begins, you may have chemotherapy first with carboplatin and paclitaxel together. This is called induction chemotherapy.

If any lymph nodes in the abdomen are known or suspected to have cancer, radiation will be given to a larger treatment area that includes these nodes as well.

Testing to find recurrence early

After finishing treatment, you will have testing to look for early signs of possible recurrence. This is known as surveillance. The information that follows applies to surveillance for common types of cervical cancer and for small cell neuroendocrine carcinoma of the cervix (NECC).

Physical exams

Physical exams performed by your oncologist are an important part of surveillance, especially in the first 5 years after treatment. At these follow-up visits, tell your provider about any changes in your health, such as new or worsening symptoms.

The recommended time frames for these follow-up visits are listed below. Because everyone's individual risk of recurrence is not the same, ranges are used. Those at higher risk of recurrence may benefit from more frequent exams than those at lower risk.

- **Years 1 and 2**: Every 3 to 6 months
- **Years 3, 4, and 5**: Every 6 to 12 months
- **After year 5**: Once a year (or as agreed upon with your cancer care provider based on your personal risk of recurrence)

Imaging

Long-term surveillance imaging generally isn't needed after treatment. Imaging is typically ordered if you have new or worsening symptoms, or if other findings suggest recurrence or spread. Follow-up imaging according to the cancer stage is listed in **Guide 3**.

Pap tests

Depending on the type of surgery you have (if any), your doctor may recommend yearly Pap screening tests after treatment. Pap testing is helpful for finding new areas of abnormal and precancerous cells. It isn't as good at detecting recurrent cervical cancer. Pap testing is also known as cervical or vaginal cytology.

Blood tests

If you develop symptoms, or if a physical exam is suspicious for recurrence, your provider may order blood tests.

In addition to a complete blood count (CBC), testing may measure blood urea nitrogen (BUN) and creatinine levels. These check your kidney function. Liver function tests may also be ordered.

Survivorship

In addition to surveillance testing, a range of other care is important for cancer survivors. This includes keeping alert for symptoms of cancer recurrence. For more information, see *Chapter 7: Survivorship*.

Recurrence

This section discusses cervical cancer that doesn't improve with treatment (persistent) or that returns after treatment (recurrent). If your provider suspects recurrence based on your symptoms or a physical exam, you will have imaging tests to check.

If needed, exploratory surgery can help guide care decisions. This procedure allows your surgeon to look inside your body and see the cancer first-hand.

Biomarker testing

Biomarkers are features of a cancer that can help guide treatment. Many are mutations (changes) in particular genes.

Biomarker testing is recommended for all recurrent cervical cancers. The results can be used to determine whether you can join certain clinical trials, and whether you may benefit from specific targeted therapies.

Some providers order biomarker testing early in the course of treatment. Others wait and order it only if the cancer returns or spreads.

Guide 3
Recommended imaging after treatment is complete

Stage 1	**After non–fertility-sparing treatment** • If you had radiation or chemoradiation, or if the cancer is stage 1B3, you may have a PET/CT scan 3 to 6 months after finishing treatment. The area from your neck to your mid-thigh is typically scanned. **After fertility-sparing treatment** • You may have an MRI of your pelvis (with contrast) 6 months after surgery, and then yearly for 2 to 3 years.
Stages 2, 3, and 4A	• Imaging with either FDG-PET/CT **(preferred)** or CT with contrast is recommended 3 to 6 months after treatment. The area from your neck to your mid-thigh will be scanned. • Other imaging may include a pelvic MRI (with and without contrast) 3 to 6 months after finishing treatment.
Metastatic cancer (stage 4B or distant recurrence)	Imaging is used mainly to learn how the cancer is responding to systemic therapy. Imaging may include CT, MRI, or PET/CT.

When possible, testing should be performed on tumor tissue removed during a biopsy or surgery. Otherwise, a sample of blood may be tested instead.

Testing for PD-L1 expression is recommended for everyone with recurrent, worsening, or metastatic cervical cancer. If the PD-L1 level exceeds a set cutoff point, the cancer is considered PD-L1-positive.

The biomarkers listed below are less common in cervical cancer but should still be tested for.

- HER2
- Mismatch repair (MMR)
- Microsatellite instability (MSI)
- Tumor mutational burden (TMB)
- *NTRK* gene fusion (for those with a cervical sarcoma)
- *RET* gene fusion

If cancer returns to the pelvis

Cervical cancer that returns to the pelvis only is known as a local or regional recurrence. If the area hasn't been treated with radiation and the cancer can be surgically removed, surgery is generally preferred. After surgery, EBRT and systemic therapy are recommended. Brachytherapy may also be given.

If the area has already been treated with radiation and it is no longer an option, possible treatment options are listed next. The best option(s) for you will depend on the specific location of the new cancer growth in the pelvis.

- Pelvic exenteration surgery
- Radical hysterectomy
- Brachytherapy
- Individualized EBRT and possibly systemic therapy
- Systemic therapy
- Resection of new cancer growth

Radical hysterectomy and brachytherapy are typically only considered in carefully selected patients. Supportive care is always an option, whether you are in treatment or not.

Pelvic exenteration

Pelvic exenteration is a radical surgery that involves removing all remaining organs of the reproductive system. This includes the uterus, fallopian tubes, ovaries, and vagina. Nearby organs, including the bladder, rectum, and anus may also be removed. The goal is to cure the cancer by removing all the organs to which cancer has or may spread.

It's often possible to reconstruct the vagina after pelvic exenteration. A surgeon can create an artificial vagina using muscle from another area of your body, often a muscle in the lower abdomen.

If the bladder or organs involved in bowel function are removed, a diversion is typically performed. Diversions are surgical procedures that redirect urine and stool to new openings through which they exit the body.

If your bladder is removed, your surgeon will use a piece of your intestine to create a tube that allows your kidneys to drain. This is called an ileal or incontinent diversion. Your urine will exit the body through a small

opening called a stoma. A small disposable bag attached to the outside of your abdomen collects the urine. This is called an ostomy bag or ostomy pouch.

Another option if your bladder is removed, is an artificial bladder or Indiana pouch. This type of diversion doesn't require an ostomy bag. In this method, your surgeon uses a piece of intestine to create a pouch inside the body to hold urine. To drain the urine, you insert a catheter through a small stoma in the abdominal wall (or sometimes the belly button).

If the rectum is removed, a permanent colostomy may be created. A colostomy connects the remaining colon to the outside of the abdomen. Stool exits the body through a stoma and goes into a bag attached to the skin.

In some cases, the remaining colon can be connected to the remaining rectum or anus. In this case, a permanent colostomy may not be required and you may retain near-normal bowel function. But if the anus is removed, a permanent colostomy is always needed.

Metastatic cancer

Cervical cancer that isn't diagnosed early enough often spreads to the liver, lungs, or bones. If this occurs, the cancer is described as metastatic.

If the cancer had already metastasized when it was found, it is stage 4B. If the cancer returns and metastasizes after treatment, it is called a distant recurrence. The following information applies to both.

Systemic therapy is recommended for metastatic cancer. It helps limit tumor growth and prevent further spread. Choosing the best systemic therapy(ies) for you begins with biomarker testing, if not already performed. Whether the cancer has the PD-L1 biomarker plays a key role in selecting a first-line regimen. Preferred options are listed in **Guide 4**.

Preferred therapies have the most evidence they may work better and .be safer than other therapies. If you can't have the preferred therapies listed in **Guide 4**, there are other recommended options for first-line treatment.

Local therapies

Local therapies are often an option for treating or controlling metastatic cancer, alone or with systemic therapy. These methods remove or destroy new cancer growth where it forms. Resection (surgery), external radiation therapy, and ablative therapies are types of local therapies.

Ablative therapies destroy areas of cancer with little harm to nearby tissue. They are often delivered using a probe or electrode placed directly into or next to the tumor.

Stereotactic body radiation therapy (SBRT) is a highly specialized type of radiation therapy that may be used to treat tumors in the liver,

Guide 4
Preferred first-line therapies for recurrent or metastatic cancer

Preferred for tumors with the PD-L1 biomarker	• Cisplatin, Paclitaxel plus Pembrolizumab (Keytruda) • Carboplatin, Paclitaxel plus Pembrolizumab (Keytruda) These regimens may also include Bevacizumab (Avastin), a biologic therapy designed to work with chemotherapy.
Preferred for all other common tumor types	• Cisplatin, Paclitaxel plus Bevacizumab • Carboplatin, Paclitaxel plus Bevacizumab • Cisplatin, Paclitaxel plus Atezolizumab (Tecentriq) plus Bevacizumab • Carboplatin, Paclitaxel plus Atezolizumab (Tecentriq) plus Bevacizumab

lungs, or bone. Only a few treatment sessions are needed.

If first-line therapy doesn't work well, or stops working

The first systemic therapy you receive may not control the growth or spread of the cancer.You have other options to try next, as second-line therapy.

Immunotherapy is the preferred second-line treatment for tumors with certain biomarkers. For all other tumors, a type of therapy called an antibody-drug conjugate is preferred. **See Guide 5.**

Preferred therapies have the most evidence they may work better and be safer than other therapies. If you can't have the preferred therapies in **Guide 5**, there are other recommended options for second-line (and later) treatment.

You may also consider enrolling in a clinical trial. Ask your treatment team if there is a clinical trial that you can join.

If immunotherapy is planned, see *NCCN Guidelines for Patients for Immunotherapy Side Effects: Immune Checkpoint Inhibitors*, available at NCCN.org/patientguidelines and on the Patient Guides for Cancer app.



Guide 5
Preferred second-line therapies for recurrent or metastatic cancer

Preferred for tumors with any of these biomarkers: • **PD-L1–positive** • **TMB-H** • **MSI-H/dMMR**	Pembrolizumab (Keytruda)
Preferred for all other common tumor types	Tisotumab vedotin-tftb (Tivdak)

Supportive care

Supportive care is an important part of cancer care. The goal is to improve your quality of life during and after cancer treatment. Supportive care is for everyone with cancer and their families, not just for those at the end of life. It's also known as palliative care.

Supportive care includes a wide range of services. Supportive care prevents or manages the symptoms of cancer and the side effects of cancer treatment, like pain and cancer-related fatigue. It also addresses the mental, social, emotional, and spiritual concerns faced by people with cancer.

Supportive care provides help with additional needs, such as:

- Making treatment decisions
- Coordinating your care
- Paying for care
- Planning for advance care and end of life

Read more about the types of support you may receive in *NCCN Guidelines for Patients: Palliative Care*, available at NCCN.org/patientguidelines and on the Patient Guides for Cancer app.



Advance care planning

Talking with your doctor about your prognosis can help with treatment planning. If the cancer cannot be controlled or cured, a care plan for the end of life can be made. Benefits of advance care planning include:

- Knowing what to expect
- Making the most of your time
- Lowering the stress of caregivers
- Having your wishes followed
- Having a better quality of life

Advance care planning starts with an honest talk between you and your doctors. Just having a general idea of your prognosis will help you decide at what point you may want to stop treatment, if at all.

Key points

- Fertility-sparing treatment is an option for some stage 1 cervical cancers, if desired. Treatment typically involves either cone biopsy or radical trachelectomy.
- Non–fertility-sparing treatment for early-stage cervical cancer often involves hysterectomy. External beam radiation therapy (EBRT) or chemoradiation may follow surgery. If you can't have surgery, EBRT and brachytherapy are recommended.
- Extra imaging or surgical staging may be done for locally advanced cancers to see if there is cancer in lymph nodes. Treatment with chemoradiation and brachytherapy is recommended for most locally advanced cervical cancers.
- Surveillance after treatment involves physical exams and staying alert for symptoms of recurrence or spread.
- Follow-up imaging is recommended 3 to 6 months after finishing treatment for stage 2, 3, or 4A cancer. After that, imaging will be ordered if you have symptoms or there are other possible signs of relapse.
- If you haven't had treatment with EBRT, it is often used to treat cancer that returns to the pelvis. If the cancer can't be treated with radiation, pelvic exenteration surgery may be an option.
- For metastatic cancer, systemic therapy helps limit tumor growth and prevent further spread. Biomarker testing is used to help choose the best regimen(s) for you. Local therapies may be used to remove or destroy new cancer growth.

Questions to ask

- If surgery is planned, am I a candidate for fertility-sparing treatment?
- What symptoms should I look out for during treatment?
- How likely is it that the cancer will return?
- Has my tumor been sent for biomarker testing? Does the cancer have any biomarkers?

6

Treatment for neuroendocrine carcinoma of the cervix

53 Early NECC

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This chapter describes treatment for small cell neuroendocrine carcinoma of the cervix (NECC). Treatment often involves chemotherapy and radiation therapy. Surgery may be an option for small cancers.

Early NECC

Neuroendocrine carcinoma of the cervix (NECC) that hasn't spread beyond the cervix is stage 1. Treatment is based on the size of the tumor.

Tumors 4 cm or smaller

Stages 1A1, 1A2, 1B1, and 1B2 cancers are 4 centimeters (cm) or smaller.

If you are able to have surgery, the preferred treatment is radical hysterectomy. Your surgeon will remove nearby lymph nodes from the pelvis (and maybe from the abdomen) for testing. After surgery, treatment with either chemotherapy or chemoradiation is recommended.

A second treatment option for tumors 4 cm or smaller is chemoradiation and brachytherapy. When treatment is over, your provider may recommend more chemotherapy.

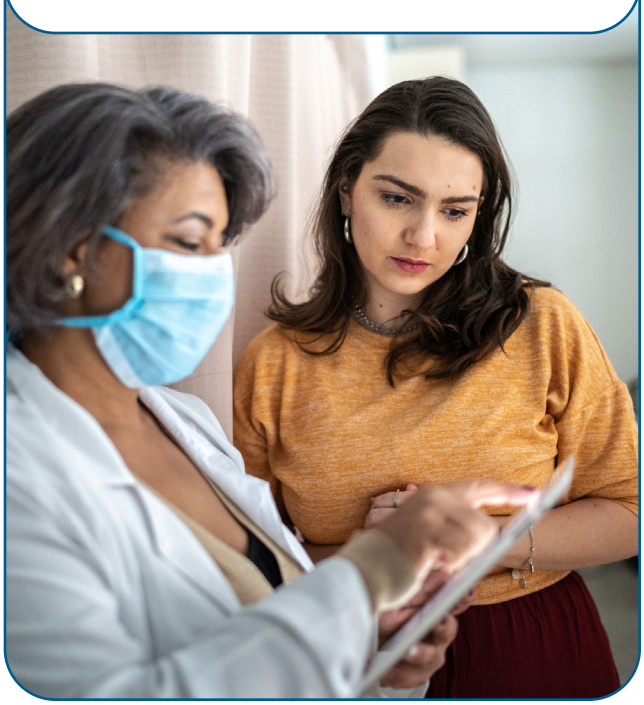
Tumors larger than 4 cm

Tumors in the cervix that are larger than 4 cm are stage 1B3. The preferred treatment plan for these tumors begins with chemoradiation and brachytherapy. More chemotherapy may follow.

Another possibility is that chemotherapy will be given first to try to shrink the cancer. If chemotherapy works well, radical hysterectomy may be your next treatment. If you have this surgery, treatment with external beam radiation therapy (EBRT) alone or chemoradiation may be given next. More chemotherapy may follow.

If radical hysterectomy isn't performed, treatment with chemoradiation and brachytherapy is the next step. More chemotherapy may follow.

Enrolling in a clinical trial allows you to receive treatment while helping researchers learn more about this rare type of cervical cancer. For more information on clinical trials, see the end of *Chapter 4: Types of treatment*.



Locally advanced NECC

Cervical cancer that has grown beyond the uterus but hasn't spread to the liver, lungs, or bone is described as locally advanced. Stages 2, 3, and 4A NECC are locally advanced. These cancers are non-metastatic.

At this time, treatment with chemoradiation and brachytherapy is preferred for locally advanced NECC. When these treatments are over, you may have more chemotherapy.

The other treatment strategy for locally advanced NECC is having chemotherapy first, followed by chemoradiation and brachytherapy.

If the selected approach works well, surveillance will begin. Surveillance involves testing to look for early signs of recurrence. The surveillance strategy for common types of cervical cancer is also recommended for NECC.

Persistent cancer

Cancer that doesn't improve with initial treatment is described as persistent. Persistent NECC is typically treated with systemic therapy.

Pelvic exenteration surgery may also be an option to discuss with your provider. This radical surgery involves removing all of the organs in the pelvis to which cancer has (or may) spread.

Supportive care is also available to everyone with cancer. This care refers to a range of resources, such as hospice care, access to pain specialists, and emotional and spiritual support.

Recurrence or spread

The cancer may return to the same general area in the pelvis after treatment. In this case, treatment options typically include:

- Systemic therapy
- Pelvic exenteration surgery

If systemic therapy is planned, you will receive one of the regimens in **Guide 6**.

NCCN experts recommend options for systemic therapy based on science and safety. When helpful, they assign a level of preference to their recommendations:

- **Preferred therapies** have the most evidence they work better and may be safer than other therapies.
- **Other recommended therapies** may not work quite as well as preferred therapies, but they can still help treat cancer.
- **Therapies used in certain cases** work best for people with specific cancer features or health circumstances.

The preferred and other recommended therapies for recurrent and metastatic NECC are listed on the next page.

Spread to liver or lungs

The cancer may return to areas far from the pelvis, such as the liver or lungs. This is called a distant recurrence. The cancer is metastatic.

Treatment with systemic therapy is preferred for metastatic NECC. It helps keep the cancer under control and prevent further spread. For your first course of chemotherapy, you will receive one of the regimens in **Guide 6**.

If the cancer doesn't improve, or returns again, there are other good options for chemotherapy. Also, checkpoint inhibitor immunotherapy is a newer option for second-line treatment of recurrent or metastatic NECC.

If immunotherapy is planned, see *NCCN Guidelines for Patients for Immunotherapy Side Effects: Immune Checkpoint Inhibitors*, available at NCCN.org/patientguidelines and on the Patient Guides for Cancer app.



Guide 6
Recurrent or metastatic NECC: Options recommended first

Preferred options	• Cisplatin, Etoposide • Carboplatin, Etoposide
Other recommended options	• Cisplatin, Etoposide plus Atezolizumab (Tecentriq) or Durvalumab (Imfinzi) • Carboplatin, Etoposide plus Atezolizumab or Durvalumab • Topotecan, Paclitaxel plus Bevacizumab • Cisplatin, Paclitaxel (if you've already received Cisplatin, Carboplatin is the recommended substitute)

Key points

- Small cell neuroendocrine carcinoma of the cervix (NECC) is a rare and usually fast-growing type of cervical cancer.
- Treatment for NECC often involves chemotherapy, external beam radiation therapy (EBRT), and brachytherapy. Surgery (along with other treatment) may also be an option for stage 1 NECC.
- Options for treating persistent or recurrent NECC that is only in the pelvis include systemic therapy and possibly pelvic exenteration surgery.
- Supportive care is an option for everyone with NECC. Talk to your care team about resources available to you.
- There is much to be learned about rare cancers like small cell NECC. Think about enrolling in a clinical trial for treatment.

Questions to ask

- Am I a candidate for surgery?
- Do you know of any available clinical trials I am eligible for?
- What are the side effects of the preferred systemic therapy regimens?



We want your feedback!

Our goal is to provide helpful and easy-to-understand information on cancer. Take our survey to let us know what we got right and what we could do better.
NCCN.org/patients/feedback

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Survivorship focuses on the physical, emotional, and financial issues unique to cancer survivors. Survivorship begins at diagnosis and continues throughout your lifetime.

Symptoms of recurrence or spread

If cervical cancer returns, it can affect your body in ways that you can feel or notice (symptoms). Symptoms that may mean cervical cancer has returned or spread include:

- Vaginal discharge or bleeding
- Blood in your urine or stool
- Loss of appetite or weight loss
- Pain in the pelvis, hips, back, or legs
- Coughing
- Shortness of breath
- Swelling in your stomach or legs

If you notice any of these symptoms, contact your care team right away. Don't wait for your next scheduled visit.

Bowel and bladder changes

After surgery or radiation therapy for cervical cancer, it can be harder to hold your urine (pee) in than before. The urge to pee may also come on more suddenly and strongly.

Your bowel movements may be watery and more frequent. You may also notice blood while peeing, or from a bowel movement.

Pelvic floor physical therapy, described next, can help with bowel and bladder changes.

Pelvic floor physical therapy

The pelvic floor is a group of muscles that supports the organs of the pelvis. These muscles play a key role in bowel and bladder control as well as sexual function and arousal.

Pelvic floor physical therapy can strengthen these muscles before and after treatment. Therapy can include at-home exercises to tighten and release the vaginal and anal muscles (Kegel exercises) as well as hands-on techniques by a physical therapist.

Ask your care team for help finding a pelvic floor specialist in your area.

Early menopause

If you are pre-menopausal, removing the ovaries or exposing them to radiation causes a sudden drop in estrogen and progesterone. This results in infertility, and possibly menopausal symptoms. Your periods may stop and you may experience hot flashes, night sweats, weight gain, and mood changes.

The lining of the vagina can become thin, dry, and inflamed. This is called vaginal atrophy. Not having enough estrogen can also have long-term risks, including heart disease and bone loss (osteoporosis).

Menopausal hormone therapy

When these hormonal changes cause symptoms of menopause, menopausal hormone therapy (MHT) may be an option. This can include systemic (oral or intravenous) estrogen (combined with progestins for those with a uterus intact) and vaginal applications of estrogen. Speaking with a specialized menopausal symptom team may be helpful to determine whether MHT is right for you.

Sexual and vaginal health

Sexual side effects can occur after cervical cancer treatment, including:

- Reduced sex drive (libido)
- Vaginal dryness
- Pain during sex
- Narrowing and shortening of the vagina (vaginal stenosis)

Vaginal moisturizers

Older age, menopause, and some cervical cancer treatments can cause the vagina to become dry and less stretchy. To offset this side effect, use of water-based vaginal moisturizers is highly encouraged.

Vaginal moisturizers restore moisture to the vagina and help to keep the vaginal tissue healthy. Vaginal moisturizers can be used daily. Many come with applicators to make

Premature menopause

If you are premenopausal, surgically removing the ovaries or exposing them to radiation can cause symptoms of menopause, including sleeping problems. Consulting with a menopause specialist is recommended.



using them easier. Vaginal estrogen cream or tablets may be particularly helpful where there has been thinning of the vaginal and vulvar tissues due to loss of estrogen.

Vaginal dilator therapy

Radiation therapy to the pelvis can cause the vagina to become shorter and narrower. This is called vaginal stenosis. Vaginal stenosis can make it uncomfortable or even painful to have sex, or to have vaginal exams by a doctor.

Using a vaginal dilator can prevent or treat vaginal stenosis. This is a device used to gradually stretch or widen the vagina. You can start using one as soon as 2 to 4 weeks after radiation therapy has ended, and continue to use it for as long as you want.

Sexual health therapists

While it may be uncomfortable to talk about sexual health, keep in mind that these side effects are common and can often be managed or lessened. Consider seeing a sexual health therapist. These health care professionals specialize in helping cancer survivors and others overcome and manage sexual side effects of cancer treatment.

Many cancer treatment centers have programs focused solely on sexual health after cancer treatment. Ask your doctor about resources available through your cancer center that can help minimize the impact of cancer treatment on your sexual health.

Mental health and wellness

It can be hard to cope with the effects of cancer and its treatment. Many survivors report having a lower quality of life after cancer treatment. Depression, anxiety, fear of recurrence, and trouble adjusting to changes in the body are possible.

Many people also have financial stressors, such as concerns or hesitation about returning to work and insurance coverage issues. Personal relationships, sexuality, and intimacy may also be affected by a cancer diagnosis or cancer treatment.

If you are anxious, distressed, depressed, or are just having trouble coping with life after cancer, you are not alone. Expect your treatment team to ask about your mental health. If they don’t, speak up. There are resources available that can help.

Social workers at your treatment center are often excellent resources to help connect you with mental health and financial resources.

Other possible side effects

Weakened bones

Radiation treatment to the pelvis can weaken bones in the pelvis, putting you at increased risk of fractures. Your doctor may want to start monitoring the density of your bones.

Nerve damage

Chemotherapy can damage the sensory nerves. This is known as neuropathy. It often begins in the hands or feet and gets worse over time. The damage can result in pain, numbness, tingling, swelling, or muscle weakness in different parts of the body. If the damage is painful, the feeling is often described as a shooting or burning pain.

Lower body swelling

Treatment for cervical cancer often involves removing lymph nodes during surgery. Lymph may not drain properly after lymph nodes are removed. This can result in lymphedema. Lymphedema is swelling caused by a build-up of lymph fluid in tissues. It most often occurs in the lower part of the body for cervical cancer survivors.

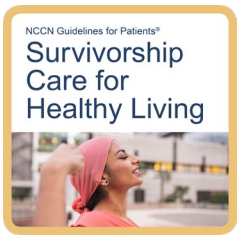
Ostomy care

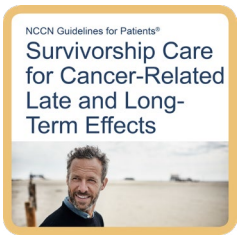
If you have an ostomy, you may want to join an ostomy support group. Another option is to see a health care provider who specializes in ostomy care, such as an ostomy nurse. People with ostomies can still live active lifestyles. Consider consulting with an ostomy professional before undertaking vigorous physical activity.

More information

For more information on cancer survivorship, the following are available at NCCN.org/patientguidelines and on the NCCN Patient Guides for Cancer app:

- *Survivorship Care for Healthy Living*
- *Survivorship Care for Cancer-Related Late and Long-Term Effects*





Healthy habits for cancer survivors



Follow your primary care provider's recommendations for screening for other cancers based on your age, sex, and risk level



Get other recommended health care such as blood pressure checks, hepatitis C screening, and immunizations (like the flu shot)



Eat a diet rich in plant-based foods. Limit red meat and processed foods.



Alcohol can increase the risk of some cancers. Drink little to no alcohol.



Move more, rest less. Try to exercise at a moderate intensity for at least 150 minutes per week.



Maintain a healthy weight. Tracking your weight, diet, calories, and activity levels may help you meet your goals.



Stop using tobacco. Ask your care team about options to help you quit. Counseling and medication are often options.



Radiation treatment to the pelvis increases the risk of pelvic fractures. Your provider may start checking your bone density.

Key points

- Survivorship care aims to improve the physical, emotional, and financial health of cancer survivors. Ask your cancer care provider for a written survivorship care plan.
- Try to stay alert for signs of potential recurrence or spread, including vaginal discharge or bleeding, blood in your urine or stool, and weight loss.
- Many survivors experience changes in their bathroom habits and sexual function after treatment. Other possible side effects include premature menopause, impaired fertility, insomnia, painful nerve damage, and swelling of the legs.
- Pelvic floor physical therapy, menopausal hormone therapy, and vaginal moisturizers and dilators can help with some side effects. Sexual health therapists help cancer survivors and others manage sex-related side effects.
- Depression, anxiety, fear of recurrence, and trouble adjusting to changes in the body are possible after cervical cancer treatment. Ask your care team about available mental health resources.

Questions to ask

- Can you give me a written survivorship care plan?
- Does this treatment center have a survivorship care program?
- What can I expect during a survivorship care visit?
- Can you connect me with support groups or other survivors?



Take our survey and help make the NCCN Guidelines for Patients better for everyone!

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Want to learn more? Here's how you can get additional help.

What else to know

This guide helps you know your options so you can make informed decisions and improve your cancer care. But it's not the only resource that you have.

Ask for as much information and help as you need. Many people are interested in learning more about:

- Managing side effects
- Building a support system
- Lifestyle changes after cancer
- Coping with other health problems
- The latest in cancer research

What else to do

Your health care center can help you with next steps. It often has on-site resources to help meet your needs and find answers to your questions. Health care centers can also inform you of resources in your community.

In addition to help from your providers, the resources listed in the next section provide support for many people like yourself.

Where to get help

Look through the list below and visit the provided websites to learn more about these organizations.

CancerCare
cancercare.org

Cancer Hope Network
cancerhopenetwork.org

Cervivor
cervivor.org

GRACE
cancergrace.org

HPV Cancers Alliance
hpvalliance.org

Imerman Angels
imermanangels.org

Ovarian Cancer Research Alliance (OCRA)
ocrahope.org

Triage Cancer
triagecancer.org

Questions to ask

- Who can I talk to about help with housing, food, and other basic needs?
- What help is available for transportation, childcare, and home care?
- Are there other services available to me and my caregivers?



Words to know

abdomen
The belly area between the chest and pelvis.

adenocarcinoma
The second most common type of cervical cancer. Starts in glandular cells in the endocervix.

adenosquamous carcinoma
A less common type of cervical cancer that contains both squamous cells and gland-like cells.

biomarker
Features of a cancer that can help guide your treatment. Many biomarkers are somatic mutations (acquired changes) in the DNA of the cancer cells.

brachytherapy
A type of radiation therapy in which radioactive material sealed in needles, seeds, wires, or catheters is placed directly into or near a tumor. Also called internal radiation therapy.

cancer stage
A rating of the extent of cancer in the body.

cervical intraepithelial neoplasia (CIN)
Abnormal and potentially pre-cancerous cells on the surface of the cervix. Also called cervical dysplasia.

cervix
The lower part of the uterus that connects to the vagina.

clinical trial
Research on an investigational test or treatment to assess its safety or how well it works.

cone biopsy
A procedure in which a cone-shaped piece of abnormal tissue is removed from the cervix. May be used to gather more information about the extent of a cancer or to treat some early cervical cancers. Also called cervical conization.

ectocervix
The rounded, outer part of the cervix that extends into the vagina. The ectocervix is lined with cells called squamous cells.

endocervix
The inner part of the cervix that forms a canal between the vagina and the uterus. The endocervix is lined with glandular cells that make mucus.

external beam radiation therapy (EBRT)
Treatment with radiation received from a large machine outside the body. Types of EBRT include intensity-modulated radiation therapy (IMRT) and stereotactic body radiation therapy (SBRT).

gynecologic oncologist
A surgeon who is an expert in cancers that start in the female reproductive organs.

human papillomavirus (HPV)
A common sexually transmitted infection. Almost all cervical cancers are caused by long-term HPV infection.

induction chemotherapy
Chemotherapy given before planned treatment with chemoradiation.

intensity-modulated radiation therapy (IMRT)
An advanced type of external radiation therapy often used to treat cervical cancer. Allows a high dose of radiation to reach the tumor while limiting the amount that reaches surrounding normal tissue and nearby organs and structures.

lymph
A clear fluid containing white blood cells that fight infection and disease.

lymph nodes
Small groups of disease-fighting cells located throughout the body.

lymphovascular space invasion (LVSI)
The presence of tumor cells in the blood vessels or lymph vessels inside the tumor. Cancers with LVSI are at higher risk of spreading than those without.

medical oncologist
A doctor who is an expert in treating cancer with systemic therapies, such as chemotherapy.

neuroendocrine carcinoma of the cervix (NECC)
A rare and often fast-growing type of cervical cancer.

neuropathy
A nerve problem that causes pain, tingling, and numbness in the hands and feet.

ovarian transposition
Surgery that moves one or both ovaries out of the range of the radiation beam. Also called oophoropexy.

ovary
One of a pair of glands that make hormones and eggs for sexual reproduction.

parametrium
The fat and connective tissue surrounding the uterus. The parametrium helps connect the uterus to other tissues in the pelvis.

pathologist
An expert in testing cells and tissue to find disease.

pelvic exenteration
A radical surgery used to treat cervical cancer that returns to the pelvis. The uterus (including the cervix), vagina, ovaries, and fallopian tubes are removed. The bladder, rectum, and anus may also be removed.

pelvis
The area of the body between the hip bones.

platinum chemotherapy
Treatment with 2 or more chemotherapy drugs in which the main drug is made with platinum.

premature menopause
The start of menopause caused by removing the ovaries or exposing them to radiation. Results from the sudden drop in estrogen in the body.

radiation oncologist
A doctor who is an expert in treating cancer with radiotherapy.

recurrence
The return of cancer after treatment. Also called a relapse.

squamous cell carcinoma (SCC)
The most common type of cervical cancer. Starts in squamous cells lining the ectocervix.

supportive care
A range of resources and care to improve quality of life for cancer survivors. Also called palliative care.

targeted therapy
Treatment with drugs that target a specific or unique feature of cancer cells.

trachelectomy
Surgery to remove the cervix. The upper part of the vagina and certain pelvic lymph nodes may also be removed.

transformation zone
The area where the endocervix and ectocervix meet. Most cervical cancers and pre-cancers start in the transformation zone. Also called the squamocolumnar junction.

type A hysterectomy
Surgery to remove only the uterus (including the cervix). Most commonly used for very early cervical cancers. Also called simple hysterectomy.

type B radical hysterectomy
Surgery to remove the uterus (including the cervix), some connective tissue that holds the cervix in place, and about a half inch or less of the vagina. Type B hysterectomy is performed through the abdomen. Also called modified radical hysterectomy.

type C1 radical hysterectomy
Surgery to remove the uterus (including the cervix), much of the connective tissue that holds the cervix in place, and the top quarter or third of the vagina. Type C1 hysterectomy is performed through the abdomen.

uterus
The organ where a fetus grows and develops during pregnancy. Also called the womb.

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This patient guide is based on the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) for Cervical Cancer, Version 2.2026. It was adapted, reviewed, and published with help from the following people:

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University of Michigan Rogel Cancer Center
Ann Arbor, Michigan

University of Wisconsin Carbone Cancer Center
Madison, Wisconsin

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